

Zero-change construals as a probe into event structure: Mandarin Chinese simple verbs and their anticausative vs. antiagentive alternants

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Abstract. Speakers of many languages can say the equivalent of *I burned the book, but it didn't burn at all*, asserting an action while denying that the change it aims at ever began. We refer to this interpretation as ZERO-CHANGE. We show that variation in its availability across syntactic frames in Mandarin simple verbs provides a novel probe into event structure. We start from the proposal that the relation between a verbal event and its result state is not fixed in the lexicon (Martin et al. 2025). This allows one and the same verb to describe either a full causation event when its subject is an agent, or a mere change when the cause is either demoted or externalized as a causer subject. We establish that zero-change construals survive in some intransitive frames but not others, thereby teasing apart two agent-demoting operations commonly conflated: the familiar CAUSATIVE/ANTICAUSATIVE alternation and what we call the AGENTIVE/ANTIAGENTIVE alternation, whose formal semantic analysis is developed here for the first time. Distinct from passives and middles, it is attested well beyond Mandarin. Zero-change construals in Mandarin arise from two distinct sources: degree semantics (whether verbs encode a degree of event realization, rather than a degree of change), and aspect (whether languages have a partitive, non-completive perfective). Taken together, these findings yield a three-way typology of Mandarin simple verbs—change-of-state, target-state-oriented activity, and pure activity—and contribute to a broader typology of how, and under which conditions, languages allow non-culminating accomplishments. This typology exactly tracks Cheng's (1989) generalization linking telicity to participation in the transitive/intransitive alternation. We explain this correlation by showing that both properties depend on whether the event provides evidence for a state property of the theme. The three verb classes differ in how this state property is determined: it is lexically encoded in change-of-state verbs, conventionally associated with target-state-oriented activities, and, in marked cases, supplied by contextual enrichment with pure activities.

1 Introduction

A typologically striking property of Mandarin Chinese is that perfective accomplishments sentences, at least those built with an (underived) simple verb used transitively with an agentive subject, allow what we will call a ZERO-CHANGE reading.¹ Under this reading, the sentence defeasibly implies but does not entail the expected change or result described by the verb (Tai 1984, Smith 1997, Talmy 2000, Chief 2008, Koenig and Chief 2008, Zhang 2018, Tham 2019, a.o.). This phenomenon, attested in a variety of typologically diverse languages including Atayal (Chen 2017b), Hindi (Singh 1991, 1994, 1998), Indonesian (Sato 2020), Korean (Park 1993, Beavers and Lee 2020), Salish (Bar-el et al. 2005, Bar-el 2005), Tagalog (Dell 1983), Samoan and Daakaka (Hopperdietzel 2020), Tamil (Paramasivam 1979, Pederson 2008), or Thai (Koenig and Muansuwan 2000), is illustrated with the Mandarin sentences in (1)-(2) built with the perfective marker *le* and the (simple, monomorphemic) verbs *shāo* 'burn' and *shā* 'kill'.² A theoretically revealing property of Mandarin, as we shall show below, is that the zero-change reading is much easier to accept with a non-gradable verb such as *shā* 'kill' when a cardinality adverbial such as *hǎojiǐcì* 'several times' is present, whereas (1), built with the gradable verb *shāo* 'burn', is acceptable even without such an adverbial (although its presence likewise facilitates the reading). We call this asymmetry the

¹Acknowledgements omitted for review. We provide literal counterparts of Mandarin examples in English, even if these literal translations are not always felicitous in English. Abbreviations used: CL: classifier, CV: complex verb, DP: determiner phrase, DUR: durative aspect, INT: interrogative, NEG: negation, PERF: perfect, PFV: perfective, PROG: progressive, RVC: resultative verbal compound, SG: singular, SV: simple verb, TS: target state, V: verb, VP: verbal phrase.

²We use SIMPLE VERB (henceforth SVs) as a shorthand for a morphologically simplex, monomorphemic verbal stem (in contrast to morphologically complex predicates such as resultative verb compounds).

CARDINALITY ADVERBIAL EFFECT.³

- (1) *Yuēhàn shāo le tā-de shū, dàn gēnběn méi shāo-zháo.*
 Yuehan burn PFV 3SG-DE book but at.all NEG.PFV burn-ignite
 ‘Yuehan burned his book, but it didn’t get burned at all.’
- (2) *Lùlù shā le Yuēhàn #(hǎojǐcì), Yuēhàn hái huó zhe.*
 Lulu kill PFV Yuehan several.times Yuehan still live DUR
 ‘Lulu killed Yuehan several times, but Yuehan is still alive.’

The first clause in (1) with the perfective accomplishment *shāo tā-de shū* VP ‘burn his book’ by default implicates that the book was successfully burnt. However, as the second clause explicitly states, it can also be used to describe a situation where the book in question did not even start burning. The first clause in (2), containing the prototypical causative verb *shā* ‘kill’, by default implicates that the subject’s repeated attempts to kill Yuehan ended up provoking his death. However, event culmination is not entailed, since the latter’s death can be explicitly denied in the subsequent clause without yielding a contradiction. The fact that perfective sentences built with SVs such as *shāo* ‘burn’, *shā* ‘kill’ or *guān* ‘close’ fail to entail the occurrence of the expected change of the theme has also been established experimentally by [van Hout et al. \(2017\)](#) and [Liu \(2018\)](#), [Liu et al. \(Forthcoming\)](#), and documented in the data gathered by [Chief \(2008\)](#) and [Chen \(2018\)](#).

The obvious question is why sentences such as (1)–(2) are felicitous in Mandarin, while their English literal translations are not. A widespread account is that English predicates such as *burn*, *open* or *kill* and their SV counterparts in Mandarin have radically different semantics. English verbs *burn*, *open* or *kill* are change-of-state verbs: they denote a set of events leading to a result state of a certain type ([Levin and Rappaport Hovav 1995](#), [Horvath and Siloni 2011](#), [Alexiadou et al. 2015](#), [Beavers and Koontz-Garboden 2020](#) a.o). The question of whether Mandarin has change-of-state SVs is quite controversial. In a long-standing tradition going back to [Tai and Chou \(1975\)](#) and [Chu \(1976\)](#) and defended by authors such as [Shi \(1988\)](#), [Talmy \(1991, 2000\)](#), [Lin \(2004\)](#) and [Chen \(2008, 2017a\)](#) among others, Mandarin SV counterparts of English causatives (i.e., change-of-state verbs used transitively) are not themselves causative. Rather, they denote a set of activities performed in order to trigger a certain state in the theme’s referent, but the state itself is not encoded by the predicate. Thus for instance, [Chen \(2017a\)](#) translates Mandarin SVs such as *guān* by ‘do.closing’ rather than by ‘close’, to convey the idea that these SVs are activity verbs devoid of change-of-state semantics.

On this view, Mandarin SVs such as *shāo* ‘burn’ or *shā* ‘kill’ pattern on the surface like transitive activity (manner) verbs of a particular subtype, namely verbs that defeasibly implicate the attainment of a target state when the event described is successfully completed ([Talmy 1991, 2000](#), [Brisson 1994](#), [Rappaport Hovav and Levin 1998](#), [Levin 2020](#), a.o). *Wash* and *wipe* are paradigmatic examples of such verbs, as illustrated in (3) for Mandarin.

³A reviewer reports more permissive judgments for zero-change continuations with *shā* ‘kill’ even in the absence of a cardinality adverbial. We take this variation seriously: judgments on zero-change continuations with simple verbs in Mandarin are known to vary across speakers. Our claim is not that a zero-change construal is categorically unavailable for every speaker in every context, but rather that with non-gradable causative SVs, it is strongly dispreferred unless independently facilitated, for instance by a cardinality adverbial. This is the pattern confirmed by [Liu’s \(2018\)](#) and [Liu et al.’s \(Forthcoming\)](#) experimental results presented in Section 3.3.3. Furthermore, under the analysis proposed in this paper, such variation is predicted since it is treated as the reflex of a general preference for stronger interpretations, as detailed in Sections 4.3 and 4.4.

- (3) *Lùlu xǐ le nèi-jiàn dàyī, dàn gēnběn méi xǐ-gānjìng.*
 Lulu wash PFV that-CL coat but at.all NEG.PFV wash-clean
 ‘Lulu washed the coat, but it is not at all clean!’

Talmy (2000: 262–263) refers to these as ‘implied fulfillment verbs’ since the verb only defeasibly implicates the state that obtains when the described activity is successful.⁴

We refer to implied-fulfillment verbs as TARGET-STATE (TS) ORIENTED ACTIVITY VERBS, to make clear that they implicate the occurrence of a TARGET STATE (in the sense of Parsons 1990: 135); that is, a reversible, transitory state, identifiable independently of the event that causes it. Talmy contrasts implied fulfillment activity verbs with what we call PURE ACTIVITY VERBS like *kiss*, which are not conceptually associated with a specific type of target state.

In sum, under this first view, all Mandarin SVs, including *shāo* ‘burn’ and *shā* ‘kill’, lack inherent change-of-state semantics, just like TS-oriented activity verbs. This is then taken to explain why they allow a zero-change construal: the action performed by the subject may fail to bring about the expected change to any degree, simply because that change is not semantically encoded by the verb itself. Variants of this core proposal can be found in Tai and Chou (1975: 52), Chu (1976: 47), Sybesma (1997: 224), Lin (2004: 53), Chen (2017a: 698) and Zhang (2018: 129). A corollary and prevailing assumption in the literature is that true change-of-state predicates (that is, predicates describing events leading to a result state) are expressed as verbal V1-V2 compounds where V1 denotes an activity and V2 a (resulting) state (Li and Thompson 1981, Lin 2004, Xie 2009 among others).

Other authors, such as Koenig and Chief (2008), do acknowledge the existence of change-of-state verbs in Mandarin. However, they assume that a change-of-state verb such as *shā* ‘kill’ denotes a set of both complete *or* incomplete killing events. This set includes not only events that culminate in the theme’s death, but also events that merely affect the theme (e.g. by wounding it) without necessarily developing into a death-causing event.⁵

We likewise defend the view that Mandarin has morphologically underived simple verbs (SVs) with change-of-state semantics. However, we divide this class into two subclasses: gradable change-of-state verbs, such as *guān* ‘close’, and non-gradable change-of-state verbs, such as *shā* ‘kill’. This contrasts with Koenig and Chief, who analyze all Mandarin causative verbs, including *shā* ‘kill’, as gradable. We further maintain a distinction between change-of-state verbs and TS-oriented activity SVs, such as *xǐ* ‘wash’.

We offer several event structure tests for distinguishing change-of-state SVs from TS-oriented activity SVs. One of them is novel and concerns the availability of the zero-change reading across different syntactic frames, in particular intransitive constructions.

We show that the zero-change reading is uniformly available with TS-oriented activity SVs, regardless of the syntactic frame in which they occur. By contrast, with change-of-state SVs, the zero-change reading is restricted to transitive frames with an animate subject and is therefore un-

⁴These verbs form a subclass of what Beavers (2010) calls potential-for-change verbs. The latter class also contains verbs of impact such as *hit*, which behave very differently than verbs like *wash* in Mandarin, as will be made clear in Section 3.1.

⁵More precisely, Koenig and Chief (2008) assume that all change-of-state verbs that license non-culminating readings encode a scalar property. Such verbs entail that a normative scalar change has occurred to some degree *d*, where *d* > 0 rather than *d* = *d_N* (*d_N* denoting the normative degree, such as death in the case of *kill*). On their analysis, *shā* ‘kill’ therefore lexically encodes a degree of change, whereas our representation of this verb does not (see (5b) below).

available in either intransitive constructions,⁶ or transitive constructions with an inanimate subject. That change-of-state verbs tend to entail rather than defeasibly implicate a change-of-state with an inanimate, non-agentive subject is a pattern observed across many languages (see Bar-el et al. 2005 on Salish languages; Beavers and Lee 2020 on Korean; Persohn 2021 on Xhosa and Nyakysa; and more generally, Demirdache and Martin’s 2015 AGENT CONTROL HYPOTHESIS).

This observation has a broader methodological consequence. We argue that this asymmetry in the availability of the zero-change reading directly reflects a fundamental difference in the event structure of TS-oriented activity SVs and change-of-state SVs, and consequently that the distribution of the zero-change reading provides a novel diagnostic for probing event structure of Mandarin simple verbs.

To make the relevant empirical pattern explicit from the outset, Table 1 summarizes the availability of the zero-change reading across the different SV classes and syntactic frames examined in this paper. This pattern will provide the empirical basis for the typology of Mandarin simple verbs proposed in this paper (see (4)).

	TRANSITIVE		INTRANSITIVE	
	animate subj. + card. adv.	animate subj. (no adv.)	inanimate subj.	(anticausative/ antiagentive)
Pure activity SVs (<i>chuí</i> ‘hammer’)	–	–	–	–
TS-oriented activity SVs (<i>xǐ</i> ‘wash’)	✓	✓	✓	✓
CoS SVs: gradable (<i>shāo</i> ‘burn’)	✓	✓	#	#
CoS SVs: non-gradable (<i>shā</i> ‘kill’)	✓	#	#	#

Table 1: Availability of zero-change construals across SV classes and syntactic frames (✓ = available; # = unavailable, ‘CoS’ = change-of-state, ‘card. adv.’ = cardinality adverbial)

Table 1 distinguishes two types of intransitive constructions. We refer to the intransitive counterpart of a change-of-state SV as an ANTICAUSATIVE, and to the intransitive counterpart of an activity SV as an ANTIAGENTIVE. In both constructions, the theme is realized as subject and no external argument is syntactically projected. The motivation for distinguishing these two intransitive constructions is developed in Sections 3.2 and 3.4.

The availability of the zero-change reading across syntactic frames—together with three further parameters: (i) conventional or lexical-semantic encoding of a target state, (ii) participation in the transitive–intransitive alternation, and (iii) the ability to form telic predicates with a quantized object—motivates the typology of Mandarin simple verbs proposed below.

- (4) a. PURE ACTIVITY SVs
chuí ‘hammer’, *mō* ‘caress (a pet)’, *qīn* ‘kiss’, *dīng* ‘gaze at’, *tuī* ‘push’, *yā* ‘press’, *zhuài* ‘drag’.
- b. TS-ORIENTED ACTIVITY VERBS
xǐ ‘wash’, *chuī* ‘blow’, *cā* ‘wipe (dust, water)’, *tuō* ‘mop’, *jiā* ‘press from both sides (a

⁶For related observations in other languages, see Lyutikova and Tatevosov (2010: 44) on Karachay-Balkar, Ikegami (1985: 274) on Japanese, Sato (2020) on Indonesian.

finger)', *qiā* 'pinch (one's arm)', *bāi* 'bend', *zhǔ* 'cook', *kǎo* 'grill', *hōng* 'blow-dry', *xiū* 'fix/mend', *bǔ* 'fix/repair'.

c. CHANGE-OF-STATE VERBS

Gradable: *shāo* 'burn', *dòng* 'freeze', *kāi* 'open', *guān* (*mén*) 'close (the door)', *sī* 'tear', *mái* 'bury', *fā* 'leaven', *rǎn* 'dye (one's hair)', *zhé* (*yígēn shùzhī*) 'break (a branch)', *jiě* 'unknot (a cravat)', *qiē* 'cut'.

Non-gradable: *shā* 'kill', *chú* 'get rid of (the tyrant)', *zhāi* 'pick (a flower)', *guān* (*shūdiàn*) 'close (the bookstore)', *sù* 'break (a plate)', *xī* 'blow out (a candle)', *jiù* 'save (a rabbit)'.

We further argue that the zero-change reading of transitive change-of-state can arise from two distinct sources: the degree argument lexically encoded by the verb or the perfective marker *le*, which we analyze as a non-completive perfective (cf. Koenig and Muansuwan 2000 on Thai; Altshuler 2014 on Hindi). This distinction yields contrasting predictions for gradable and non-gradable change-of-state SVs-predictions that we show are borne out by the data in Section 3.3.

The paper is organized as follows. Section 2 introduces the lexical semantic assumptions adopted throughout the paper. In particular, it presents the lexical representations of activity and change-of-state SVs that underlie the semantic analysis developed in Section 4.1. Section 3 develops the proposed typology of Mandarin SVs. We first distinguish TS-oriented activity SVs from pure activity SVs and discuss the status of the target state associated with the former, as well as the default inference that this target state obtains. We then characterize the intransitive construal of activity SVs, which we term antiagentive, distinguishing it from both passives and middles. Finally, we distinguish gradable from non-gradable change-of-state SVs and introduce the anticausative alternation of change-of-state SVs. Section 4 develops the semantic analysis. We first introduce a new semantics for Mandarin change-of-state verbs and then spell out the degree semantics of gradable change-of-state verbs. Equipped with this semantics, we derive the conditions under which the zero-change construal is licensed across verb classes and syntactic frames. Section 5 provides further support for the distinction between TS-oriented activity and change-of-state SVs by introducing a new event-structure diagnostic: the availability of zero-change construals under completive modification. It also shows that modal analyses of zero-change construals (Martin and Schäfer 2017; Beavers and Lee 2020) do not readily extend to Mandarin. Section 6 examines the relation between intransitive alternations and telicity. We show that the ability to appear in an intransitive construction correlates perfectly with the ability to form a telic predicate, a correlation first observed by Cheng (1989). We derive Cheng's generalization from a state-property requirement on intransitive construals, in which sentence-final *le* plays a key role.

2 Lexical semantics assumptions

The full semantic analysis is developed in Section 4. Here we introduce the underlying lexical semantic assumptions. Following a Distributed Morphology approach (Halle and Marantz 1993), we assume that verbs are formed by combining a root with a verbalizing head (*v*).

Activity and change-of-state verbs are built with distinct verbalizing heads (Folli and Harley 2005, Folli and Harley 2007), v_{do} for activity verbs and v_{cos} for change-of-state verbs. As we argue throughout the paper, this lexical distinction underlies the contrasting distribution of the zero-change construal.

We adopt Martin et al.'s (2025) semantics for v_{cos} . As detailed in 4.1, v_{cos} introduces a relational argument *R* between events and states (5c), yielding an event predicate whose interpretation depends on the particular event-state relation supplied (5d). Depending on the syntactic

environment, R resolves as either cause or become.

- (5) a. *Lùlù shā le Mǎlā.*
 Lulu kill PFV Marat
 ‘Lulu killed Marat.’
 b. *Mǎlā shā le.*
 Marat kill PFV
 Literally: ‘Marat killed.’ [i.e. has been killed]
 c. $v_{cos} \rightsquigarrow \lambda P \lambda R \lambda e. \exists s (R(e, s) \wedge P(s))$
 d. $[v_{cos}P \text{ shā mǎlā}] \rightsquigarrow \lambda R \lambda e. \exists s (R(e, s) \wedge \mathbf{dead}(s) \wedge \mathbf{theme}(s, \mathbf{marat}))$

Both pure and TS-oriented activity SVs are formed with v_{do} . Applied to the root, the verbalizing head v_{do} returns an activity predicate (6c/d).

- (6) a. *Lùlù xǐ le shàngyī.*
 Lulu wash PFV coat
 ‘Lulu washed the coat.’
 b. *shàngyī xǐ le.*
 coat wash PERF
 ‘The coat has washed.’
 c. $v_{do} \rightsquigarrow \lambda P \lambda e. (P(e) \wedge \mathbf{act}(e))$
 d. $[v_{do}P \text{ xǐ shàngyī}] \rightsquigarrow \lambda e. (\mathbf{wash}(e) \wedge \mathbf{act}(e) \wedge \mathbf{theme}(e, \mathbf{the-coat}))$

Predicates headed by v_{do} denote activities, whereas predicates headed by v_{cos} lexically encode a relation between an event and a state. While activity predicates conceptually entail an agent—since all events in their extension are activities—they do not lexically encode a state. By contrast, predicates headed by v_{cos} do not entail an agent at the vP level. As we show in Section 4, this difference is what underlies the contrasting behavior of the two verb classes with respect to the zero-change construal.

Inspired by Piñón’s (2008) analysis of incremental theme verbs, we further assume that gradable change-of-state SVs encode a DEGREE OF EVENT REALIZATION d_e (Section 4.3). We distinguish this notion from the DEGREE OF CHANGE d_c encoded by degree achievements (Kennedy and Levin 2008), which measures a change in the extent to which an individual possesses a certain gradable property. The idea that gradable verbs may encode a degree of event realization d_e rather than a degree of change d_c is crucial for our analysis because it allows a causation event involving an agent—for example, Momo burning a book—to be realized to a positive degree without the theme undergoing any change at all (e.g., without the book beginning to burn).

3 A new typology of simple verbs in Mandarin

3.1 TS-oriented activity SVs vs. pure activity SVs

We begin by motivating the distinction between PURE ACTIVITY SVs such as *qīn* ‘kiss’ or *mō* ‘caress’ in (4a) and TS-ORIENTED ACTIVITY SVs such as *xǐ* ‘wash’ or *xiū* ‘fix’ in (4b). Unlike the latter, pure activity verbs like *kiss* are not conventionally associated with a specific target state. Consequently, when an adverb such as *successfully* modifies a pure activity verb, the resulting sentence is typically not interpreted as entailing a specific type of state. For instance, *John kissed the princess successfully* does not, in a default context, imply a state of awokeness. As Kou and Ho-

henstein (2022: section 5.1) emphasize, what distinguishes the two classes is not the availability of contextually salient goals, but rather the stability across contexts of the target state associated with the verb. With TS-oriented activity verbs, the associated target state remains constant regardless of the context and is preserved even with generic themes: *He washed the thing* still conveys a cleaning goal. By contrast, the goals associated with pure activity verbs are pragmatically inferred, need not involve a state at all, and do not carry over with generic themes (*He hit the thing* conveys no particular goal at all).

For Talmy, the inference triggered by TS-oriented activities that the target state obtains (e.g., in (3), the inference that the coat is clean(er)) is a LEXICALIZED IMPLICATURE: an inference that is both defeasible and encoded in the lexical entry of the verb. This hybrid category is problematic, however. The inference cannot be a conventional implicature, since it is defeasible. Nor can it be a scalar implicature derived from competition with a stronger form entailing the result (e.g., *clean*). If the activity were the weaker alternative, its use should then implicate that the result was *not* attained, contrary to fact. Positing a *sui generis* class of implicatures that is neither conventional nor scalar, but nevertheless ‘encoded in the lexicon’, is thus essentially an *ad hoc* move. Moreover, if the term ‘lexicalized’ is taken seriously, then the target state itself should form part of the verb’s lexical semantic representation. This is precisely the analysis standardly assumed for DEFEASIBLE CAUSATIVES, i.e. verbs that lexically encode a result state whose realisation is nevertheless defeasible in some contexts because it falls within the scope of a lexically encoded modal operator (Koenig and Davis 2001, Beavers 2011, Martin and Schäfer 2012). See (7a) and the semantic representation in (7b), following Martin and Schäfer’s (2012) proposal that defeasible causatives denote relations between an individual y , an event e , and a world w , such that e is an event whose theme is y and, in every world w' in the modal base ρ accessible from w , e causes a result state s whose theme is y .

- (7) a. Khadija explained the puzzle to Tom, but he did not understand it.
 b. $\lambda e \lambda w. \text{agent}(e, \text{khadija}) \wedge \text{explain}_w(e) \wedge \text{recipient}(e, \text{tom}) \wedge \text{theme}(e, \text{the-puzzle}) \wedge$
 $\Box_{\rho_w}^{w'} \exists s (\text{cause}_{w'}(e, s) \wedge \text{understand}_{w'}(s) \wedge \text{theme}(s, \text{the-puzzle}) \wedge \text{experiencer}(s, \text{tom}))$
 c. (Reading) this book explained the puzzle to Tom, #but he did not understand it.

Mandarin TS-oriented activity verbs, however, do not pattern with defeasible causatives. Three diagnostics support this conclusion. First, whereas most defeasible causatives do not participate in the causative alternation (Martin forthcoming), Mandarin TS-oriented activity verbs do (see Section 3.2). Second, defeasible causatives readily license inanimate (causer) subjects across languages. By contrast, TS-oriented activity verbs accept inanimate subjects far less easily. This is expected if they are activity rather than change-of-state verbs. Flexibility with respect to subject type is a hallmark of change-of-state verbs (Levin and Rappaport Hovav 1995, Alexiadou et al. 2015), whereas activity verbs are more restrictive in this regard. When activity verbs do admit inanimate subjects, these are typically interpreted not as genuine event-denoting causers, but as weak agents or effectors.⁷ Third, whenever an inanimate subject is licensed—sometimes with a

⁷We use the term *agent* in a broad grammatical sense, rather than as a synonym for an animate intentional agent. Following many authors, we distinguish agent subjects from causer subjects. With Martin et al. (2025), we assume that causer subjects denote eventualities, situations or facts and are introduced by Causer Voice. Agent subjects, by contrast, are introduced by Agent Voice and need not be animate. Thus, in examples such as English *The stone hit the window*, the inanimate DP is still construed as the effector of the action described by the verb, not as an event- or situation-denoting causer. This is the sense in which some activity predicates may admit inanimate agents. On inanimate agents in the grammar, see Cruse (1973), Piñón (2001), Fauconnier (2012), Folli and Harley (2008), Joo et al. (2025) among others.

personification flavor, as in (8b)—the two classes exhibit exactly opposite behavior with respect to the zero-change reading. Defeasible causatives typically lose this reading with inanimate subjects ((7c)), whereas TS-oriented activity verbs retain it ((8)-(9)). If Mandarin TS-oriented activities lexically encoded their target state in the same way as English defeasible causatives, the contrast between (7) on the one hand and (8)-(9) on the other would be unexpected.

- (8) a. *Lǐsì xǐ le chē-chuāng, dàn hái shì nàme zāng.*
 Lisi wash PFV car-window but still be that dirty
 ‘Lisi washed the car window, but it is still as dirty as before.’
 b. *Yǔ xǐ le chē-chuāng, dàn hái shì nàme zāng.*
 rain wash PFV car-window but still be that dirty
 ‘The rain washed the car window, but it is still as dirty as before.’
- (9) a. *Lìlǐ kǎo le ròu, dàn hái shì nàme shēng.*
 Lili grill PFV meat but still be that raw
 ‘Lili grilled the meat, but it is still as raw as before.’
 b. *Huǒ kǎo le ròu, dàn hái shì nàme shēng.*
 fire grill PFV meat but still be that raw
 ‘The fire grilled the meat, but it is still as raw as before.’

We therefore propose an alternative analysis of the default ‘target-state inference’ associated with TS-oriented activity verbs. Rather than being encoded in the verb’s lexical semantics, this default inference is derived through abductive reasoning—that is, a defeasible inference to the best explanation (Gyarmathy and Altshuler 2020). For (3), the reasoning proceeds as in (10):

- (10) a. OBSERVATION: the assertion that Lulu washed the coat.
 b. THEORY: if the coat is clean(er), then it was washed (states of cleanness conventionally come about through washing events).
 c. ABDUCED EXPLANATION: the coat is clean(er).

Since the target state is reached via non-monotonic reasoning rather than entailed or lexically encoded, its denial in the subsequent discourse is correctly predicted to be felicitous, as in (3). By contrast, pure activity verbs like *qīn* ‘kiss’ or *mō* ‘caress’, are not conventionally associated with any particular state in the relevant background theory: no conventional rule links kissing or caressing events to a specific state of the theme, and consequently no target-state inference can be derived abductively. This accounts for the asymmetry between TS-oriented and pure activity verbs. In Section 6, we show that the target-state inference is also what enables TS-oriented activity verbs (i) to form telic predicates with quantized objects and (ii) to enter the intransitive (antiagentive) construction without any contextual support, unlike pure activity SVs.

3.2 The antiagentive alternant of TS-oriented activity SVs

Another typologically striking property of Mandarin is that not only change-of-state (causative) transitives, but also activity (non-causative) transitive verbs can appear in intransitive sentences whose subject corresponds to an internal argument (Tai 1984, Cheng 1989, Aldridge 2015), as illustrated for instance in (11b) and (12b), built with the TS-oriented activity SVs *xǐ* ‘wash’ and *bǔ*

‘fix/mend’.⁸

- (11) *Shàngyī xǐ le.*
coat wash PERF
‘The coat has washed.’

- (12) *Shàngyī bǔ le.*
coat mend PERF
‘The coat has mended.’

In English, only transitive verbs with change-of-state semantics may undergo the transitive-intransitive alternation (Smith 1970, Levin and Rappaport Hovav 1995, Schäfer 2008, Alexiadou et al. 2015, a.o); accordingly, the English translations of these examples are all infelicitous.

Some authors use the label ANTICAUSATIVE for any freely alternating verb whose intransitive use takes the theme as its sole argument, regardless of whether the alternating verb has causative semantics (see, e.g., Carvalho 2016). We do not adopt this view because the label ‘anticausative’ suggests the demotion of an external *cause*, and therefore presupposes a causative, change-of-state semantics. The verbs in (11), however, are activity verbs devoid of change-of-state semantics. We reserve the label anticausative for the intransitive use of causative/change-of-state verbs (Section 3.4), and refer to the intransitive use of non-core (activity) transitive verbs such as *xǐ* ‘wash’ antiagentive. Unlike anticausatives, antiagentives denote the patient-oriented component of an activity. For instance, when used intransitively, *xǐ* ‘wash’ denotes the patient’s undergoing immersion, rubbing, movement in liquid, or a comparable process—that is, the ‘getting-washed’ component of a washing event. This component is an activity. We represent this explicitly by analyzing it as a $v_{do}P$: the **act** predicate introduced by v_{do} (recall (6a)) characterizes the event as an activity. Crucially, however, the denoted event contains no predicate relating an agent to an event, but only a theme predicate. Consequently, the antiagentive denotes the patient-oriented component of an activity. The lexical representation of intransitive *xǐ shàngyī* ‘wash the coat’ is therefore given in (13).

- (13) $\llbracket xǐ shàngyī \rrbracket \leadsto \lambda e.(\mathbf{wash}(e) \wedge \mathbf{act}(e) \wedge \mathbf{theme}(e, \mathbf{the-coat}))$

For this reason, adverbs targeting the agent of the activity are excluded. For instance, modifiers such as *shǐjìn de* ‘vigorous(ly)’, *gùyì de* ‘intentional(ly)’, can felicitously apply to a TS-oriented activity SV like *xǐ* ‘wash’ on its transitive use, as in (14), but not on its intransitive use, as shown in (15).

- (14) *Lulu shǐjìn de/ gùyì de xǐ le nèi-jiàn tīxù.*
Lulu vigorously intentionally wash PFV that-CL T-shirt
‘Lulu washed the T-shirt vigorously/intentionally.’

⁸In the intransitive examples discussed here, *le* appears in sentence-final position. In this position, *le* can in principle be verbal *le* or sentence-final *le*. Roughly, we take verbal *le* to be a perfective marker and sentence-final *le* to be a (stativizing) perfect, in line with Li et al. (1982) and Tsai (2013) among others. We assume that the *le* occurring in Mandarin antiagentive constructions is the latter: a (stativizing) perfect operator. This assumption is defended in Section 6, where we argue that sentence-final *le* plays a crucial role in making a state property available for otherwise stateless activity predicates. See Chao (1968), Teng (1973), Chan (1980), Li and Thompson (1981: 296), Sybesma (1999: 65) and Paul (2015: 14) for discussion on the semantics of sentence-final *le*.

- (15) **Nèi-jàn tīxù shǐjìn de/ gùyì de xǐ le.*
 that-CL T-shirt vigorously intentionally wash PERF
 Intended: ‘The T-shirt has washed vigorously/intentionally.’

In this respect, antiagentives differ from *bèi*-passives: the latter introduce an implicit external argument that can be targeted by agent-oriented modification, as shown in (16). The crucial difference is thus not whether an agent is conceptually present, but whether it is introduced as a syntactically represented external argument.

- (16) *Chē bèi gùyì xǐ le.*
 car BEI intentionally wash PFV
 ‘The car was washed intentionally.’

Antiagentive construals of activity verbs such as *xiě* ‘write’ or *chī* ‘eat’ are likewise incompatible with agent-oriented modifiers like *xiǎoxīn* ‘cautious(ly)’. The examples in (17) below become acceptable only when *bèi* is added.

- (17) a. *Xìn (*xiǎoxīn de) xiě le.*
 letter cautious DE write PERF
 ‘The letter wrote (cautiously).’
 b. *Xīguā (*xiǎoxīn de) chī le.*
 watermelon cautious DE eat PERF
 ‘The watermelon has eaten (cautiously).’

We propose that the above observations follow from the absence of the Voice layer that introduces the external argument (Kratzer 1996, Alexiadou et al. 2015 a.o.). In antiagentives (see (13)), as in anticausatives (Schäfer 2008), the theme is the only syntactically projected argument of the verbal predicate. Antiagentives are therefore unaccusative in the sense that they lack a Voice-introduced external argument, unlike passives (Bruening 2013).

While no VoiceP is projected in antiagentives (unlike in passives), the agent in antiagentives remains nevertheless conceptually present (as in passives). This follows from the lexical semantics of TS-oriented activity SVs: activities conceptually entail the existence of an initiator/agent, whether animate or inanimate, by virtue of containing the **act** predicate contributed by the v_{do} (cf. (13)).

That TS-oriented activity verbs retain a lexical agentive implication in antiagentive construals is shown by a diagnostic due to Mauner and Koenig (2000): if the continuation explicitly denies that anything or anyone performed the relevant activity, the result is contradictory, see (18a/b). Correlatively, as (18a/b) also show, TS-oriented activity SVs are not compatible with ‘by itself’ when used intransitively, what also confirms that the intransitive of these verbs is not an anticausative (on the *by itself* test as diagnosing anticausatives, see Chierchia 2004; Schäfer 2007; Koontz-Garboden 2009; Alexiadou et al. 2015).

- (18) a. *Shàngyī xǐ le, #dàn bú shì shéi/shénme dōngxī xǐ de, (#shì zìjǐ xǐ de)*
 coat wash PFV but NEG be who/what thing wash DE be itself wash DE
 ‘The coat washed, but it was not anyone/anything who washed it, it washed by itself.’
 b. *Biǎo xiū le, #dàn bú shì shéi/shénme dōngxī xiū de, (#shì zìjǐ xiū de)*
 watch fix PFV but NEG be who/what thing fix DE be itself fix DE.
 ‘The watch fixed, but it was not anyone/anything who fixed it, it fixed by itself.’

In contrast, ordinary anticausatives do not carry such a lexical agentive implication. The change they denote is not the patient-oriented subcomponent of an action. Consequently, explicitly denying the presence of any agent or effector does not yield a contradiction, as shown in (19). As the same example illustrates, anticausatives are compatible with *zìjǐ* ‘by itself’.

- (19) *Mén guān le, dàn bú shì shéi/shénme dōngxī guān de, shì zìjǐ guān de.*
 door close PFV but NEG be who/what thing close DE be itself close DE.
 ‘The door closed, but it was not anyone/anything who closed it, it closed by itself.’

Antientatives are impossible in languages like English. We assume that, unlike Mandarin, English does not allow VoiceP to be absent with activity transitives such as *wash*.

3.2.1 Antiagentive vs. middle construals

One might wonder whether antiagentives should be analyzed as MIDDLES.⁹ Indeed, [Maurer and Koenig \(2000\)](#) analyze the intransitive variant of English verbs like *sell*, illustrated in (20), as middles. At first glance, these constructions are very similar to Mandarin antiagentives: both lack an overt external argument while nevertheless entailing the existence of an agent, as shown by the contradictory continuation in (20) (confirmed experimentally in [Maurer and Koenig 2000](#)). Moreover, middles have been argued to lexically entail an agent that remains syntactically inaccessible ([Ackema and Schoorlemmer 2005](#), [Schäfer 2007](#)), apparently paralleling antiagentives.

- (20) The donated antique vase sold immediately (#yet no one sold it).

[Bruening \(2024a\)](#), however, argues that the agent of a middle remains accessible at some level of representation, even if not syntactically accessible. Specifically, he argues that the external argument of a middle is present in the semantics (as an index introduced by the head Voice). One argument is the compatibility of middles with instrumental phrases ([Keyser and Roeper 1984](#)), as illustrated in (21).

- (21) This drawer opens with a key. ([Bruening 2024b](#): 3)

According to Bruening, instruments are adjoined to Voice, which he assumes to be projected in middles, although the external argument itself is not. Instrumental phrases are likewise licensed in Mandarin middles without any special constraints, as illustrated in (22), where the instrumental phrase modifies the generic middle construal.

- (22) *Máo yī yìbān dài shǒutào xǐ.*
 wool.sweater usually wear glove wash
 ‘Wool sweaters usually wash with gloves on.’

⁹We thank [removed for review] for prompting us to compare antiagentive and middle constructions.

As in English, by contrast, instruments are excluded from anticausatives. The anticausative use of *shā* ‘kill’ in (23), with the theme *Lāsīpǔjīng* ‘Rasputin’ realized as subject and no external argument projected, becomes ungrammatical as soon as the instrumental phrase *yòng qiāng* ‘with a gun’ is added.

- (23) *Lāsīpǔjīng* (**yòng qiāng*) *shā le*.
 Rasputin use gun kill PERF
 Intended: ‘Rasputin killed with a gun.’

In antiagentives, the acceptability of an instrumental phrase depends on whether it can be construed as modifying the patient-oriented activity denoted by the antiagentive VP. When it can, the instrument is licensed, since it is interpreted as a participant in the event *e*. This is the case in (24): the fabric softener in (24a) and the strong glue in (24b) naturally characterize the component of the activity involving the theme, not the agent’s participation in it, rendering the instrumental phrase fully acceptable.

- (24) a. *Máoyī yòng róushùnjì xǐ le*.
 wool.sweater use softener wash PERF
 ‘The wool sweater has washed with softener.’
 b. *Chētāi yòng qiánglijiāo bǔ le*.
 tire use super.glue mend PERF
 ‘The tire has patched with super glue.’

If, on the other hand, the instrument qualifies the manner of acting, it is no longer compatible with the antiagentive, as shown in (25).

- (25) a. **Máoyī yòng shǒutào xǐ le*.
 sweater use glove wash PERF
 Intended: ‘The sweater has washed with gloves.’
 b. **Chētāi yòng zhuīzi bǔ le*.
 tire use awl mend PERF
 Intended: ‘The tire has mended with an awl.’

In these cases, the instrument necessarily characterizes the agent’s participation in the activity. Such agent-oriented instrumental PPs therefore require the Voice-related semantic index that keeps the external argument accessible, even though it is not overtly realized—an index that, following Bruening (2024a), we assume is present in middles, but which we argue is absent in antiagentives. We therefore conclude that antiagentives are distinct from middles, as the contrast between (22) and (25a) clearly demonstrates.

3.2.2 The zero-change reading in antiagentive construals

A central contribution of this paper is the use of a novel diagnostic to probe the event structure of SVs: the availability of the zero-change construal in critical contexts.

Antiagentives pass this diagnostic. In (26)-(27), for instance, the first clause remains compatible with the denial in the second clause that the conventionally associated change occurred.

- (26) *Shàngyī xǐ le, dàn gēnběn méi xǐ-gānjìng.*
 coat wash PERF but at.all NEG.PFV wash-clean
 ‘The coat has washed, but it didn’t get clean at all!’
- (27) *Shàngyī bǔ le, dàn hái gēn yuánlái yíyàng pò.*
 coat mend PERF but still as before same damaged
 ‘The coat has mended, but it is still as damaged as before!’

TS-oriented activity verbs thus license the zero-change construal in every syntactic frame examined so far: transitive frames with both animate and inanimate subjects, as well as intransitive frames. This is expected under our analysis since these verbs lexically encode only an activity (recall (13)): denying the occurrence of any change towards the TS does not contradict any component of their lexical semantics.

The distribution of zero-change construals summarized in Table 1 (Section 1) shows, however, that change-of-state SVs exhibit a markedly different pattern. The zero-change construal is unavailable in intransitive frames for all change-of-state SVs, while in transitive frames, its availability further depends on whether the verb is gradable or non-gradable. Capturing this distribution thus requires distinguishing two subclasses of change-of-state SVs — gradable causatives such as *shāo* ‘burn’ and non-gradable causatives such as *shā* ‘kill’. It also requires distinguishing change-of-state SVs from TS-oriented activity SVs, which pattern differently with respect to zero-change construals across all syntactic frames. We take up these two issues in Sections 3.3 and 3.4, respectively.

3.3 Gradable vs. non-gradable change-of-state SVs

3.3.1 Degree modification

The distinction between gradable and non-gradable change-of-state SVs is established here on the basis of their compatibility with degree expressions. Examples (28)–(31) show the distribution of several degree adverbs across the two verb classes. The same contrast can be replicated with *yíbùfēn* ‘partly’ and *yídiǎn* ‘a little’.

- (28) *Nèi-běn shū Mòmo quán shāo le.*
 that-CL book Momo complete burn PFV
 ‘Momo completely burned the book.’
- (29) #*Nèi-zhī yāzi Mòmo quán shā le.*
 that-CL duck Momo complete kill PFV
 Intended: ‘Momo completely killed that duck.’
- (30) *Nèi-běn shū Lùlu shāowēi shāo le yíxià.*
 that-CL book Lulu slightly burn PFV a.little
 ‘Lulu slightly burned the book.’
- (31) #*Nèi-tóu niú Lùlu shāowēi shā le yíxià.*
 that-CL ox Lulu slightly kill PFV a.little
 Intended: ‘Momo slightly killed that ox.’

To understand the distribution of these degree expressions, we must first determine whether

they themselves entail a positive degree of change. This is the case of *quán* ‘completely’ and *yíbùfēn* ‘partly’. Accordingly, the zero-change continuation in (32) becomes infelicitous when either of these adverbs is added.

- (32) *Nèi-běn shū Lùlu shāo le (#yíbùfēn), dànshì shū gēnběn méi shāo-zháo.*
 that-CL book Lulu burn PFV a.part but book at.all NEG.PFV burn-ignite
 ‘Lulu (partly) burned the book, but the book did not get burned at all.’

We take this to indicate that *yíbùfēn* and *quán* measure the extent to which the theme is affected (or effected) with respect to the relation *R* expressed by the verb (but not all degree expressions behave in this way, as will be shown in Section 4.3).

With these diagnostics in place, we now turn to the conditions under which gradable vs. non-gradable change-of-state SVs license zero-change construals.

3.3.2 Licensing the zero-change construal of change-of-state verbs

An overlooked and, to our knowledge, unexplained fact concerns the contexts licensing zero-change construals with animate subjects and NON-GRADABLE change-of-state verbs (e.g. *shā* ‘kill’), as opposed to GRADABLE change-of-state verbs (e.g. *shāo* ‘burn’). The relevant observation is that Mandarin speakers readily accept zero-change readings with non-gradable change-of-state SVs *only* when the verb is modified by either a cardinality adverbial (e.g. *hǎojiǐcì* ‘several times’, *yíci* ‘once’) or a durative adverbial (set aside here for reasons of space). By contrast, gradable change-of-state SVs admit the zero-change reading even without a cardinality (or durative) adverbial, although the presence of such an adverbial facilitates the denial of the entire change. We refer to this asymmetry as the cardinality adverbial effect, illustrated in (33)–(34), and in (1)–(2) above.

- (33) THE CARDINALITY ADVERBIAL EFFECT
Gōngjué shā le Lāsīpǔjīng #(hǎojiǐcì), Lāsīpǔjīng dōu méi sǐ.
 prince kill PFV Rasputin several.time Rasputin DOU NEG.PFV die
 ‘The prince killed Rasputin (several times), but Rasputin didn’t die.’
- (34) *Lìsì shāo le yì-gēn mùtōu (hǎojiǐcì), dàn gēnběn méi shāo-zháo.*
 Lisi burn PFV one-CL wood several.time but at.all NEG.PFV burn-ignite
 ‘Lisi burned a piece of wood (several times) but it didn’t get burned at all.’

A further generalization is that the zero-change construal is unavailable for change-of-state SVs in the absence of an agentive subject. This is illustrated in (35) with the gradable change-of-state verbs *guān* ‘close’ and *shāo* ‘burn’.

- (35) a. *Yí-zhèn fēng guān le nà-shàn mén, #dàn mén yìdiǎn dōu méi guān-shàng.*
 one-CL wind close PFV that-CL door but door a.little DOU NEG.PFV close-up
 Intended: ‘A gust of wind closed that door, but it didn’t get closed at all.’
- b. *Tiānqì yǎnrè, sēnlín dàhuǒ shāo le tā jiā de línzi, #kěshì línzi ānrǎnwúyǎng.*
 weather hot forest big.fire burn PFV 3SG home DE forest but forest
 untouched
 Intended: ‘It’s very hot. The forest fire burned his forest, but (his) forest was untouched.’

The same holds for the non-gradable change-of-state SV *shā* ‘kill’ in (33): the zero-change reading is felicitous only in the presence of an agentive subject. Consequently, when the subject is inanimate, denying death no longer becomes possible, and adding a cardinality adverbial cannot rescue the zero-change construal. On the contrary, it renders the sentence ill-formed, as illustrated in (36), since one can only die once.

- (36) *Shāchóngjì shā le nèi-zhī zhāngláng (#hǎojǐcì).*
 insecticide kill PFV that-CL cockroach several.time
 ‘The insecticide killed that cockroach (several times).’

The next section presents Liu’s (2018) and Liu et al.’s (Forthcoming) experimental evidence for both the Cardinality Adverbial Effect and the agentivity requirement on the zero-change construal.

3.3.3 Liu’s (2018) and Liu et al.’s (forthcoming) experimental confirmation

Liu (2018), Liu et al. (Forthcoming) report truth-value judgment tasks that test the availability of zero-change construals in (child and adult) Mandarin. Fifty adult (highly educated) native speakers from southern and northern China participated. The design manipulated two within-participant factors: CHANGE (zero-change vs. full-change) and SUBJECT TYPE (animate agent vs. inanimate causer subject). Five gradable and three non-gradable causative SVs (listed in (37)) were tested, though GRADABILITY itself was not a variable.

- (37) a. Gradable change-of-state SVs: *mái* ‘bury’, *zhé* ‘cut’, *jiě* ‘untie’, *guān* ‘close’, *kāi* ‘open’
 b. Non-gradable change-of-state SVs: *shā* ‘kill’, *sù* ‘break (a plate)’, *xī* ‘extinguish’.

In Experiment 1 (N=30), participants in the ZERO-CHANGE/AGENT condition watched videos in which an agent made a single unsuccessful attempt to perform an action (e.g. close a window, but the window does not budge). In the FULL-CHANGE/AGENT condition, the agent’s action was successful (the window was closed). Participants then answered, on the basis of the depicted events, *yes/no* questions containing perfective causative SVs without a cardinality adverbial.

By contrast, in Experiment 3 (N=20), the agent repeatedly attempts to perform the action, without any success under the ZERO-CHANGE condition, but successfully under the FULL-CHANGE condition. Participants answered, on the basis of the depicted events, *yes/no* questions containing perfective causative SVs, this time modified by the cardinality adverb *hǎojǐcì* ‘several times’. Representative test items for Experiment 1 and 3 are illustrated in (38) and (39), respectively.

- (38) *Hǎidào guān-le nà shàn mén ma?*
 Pirate close-PFV that CL door INT?
 ‘Did the pirate close that door?’
 (39) *Hǎidào guān-le hǎojǐcì nà shàn mén ma?*
 Pirate close-PFV several.times that PFV door INT?
 ‘Did the pirate close that door several times?’

Liu’s findings, summarized in Figure 1, provide clear experimental support for both the cardinality adverbial effect and the agentivity requirement. When a change-of-state SV with an animate subject is not modified by a cardinality adverbial, acceptance of perfective sentences under the ZERO-CHANGE condition is substantially higher with gradable than with non-gradable SVs

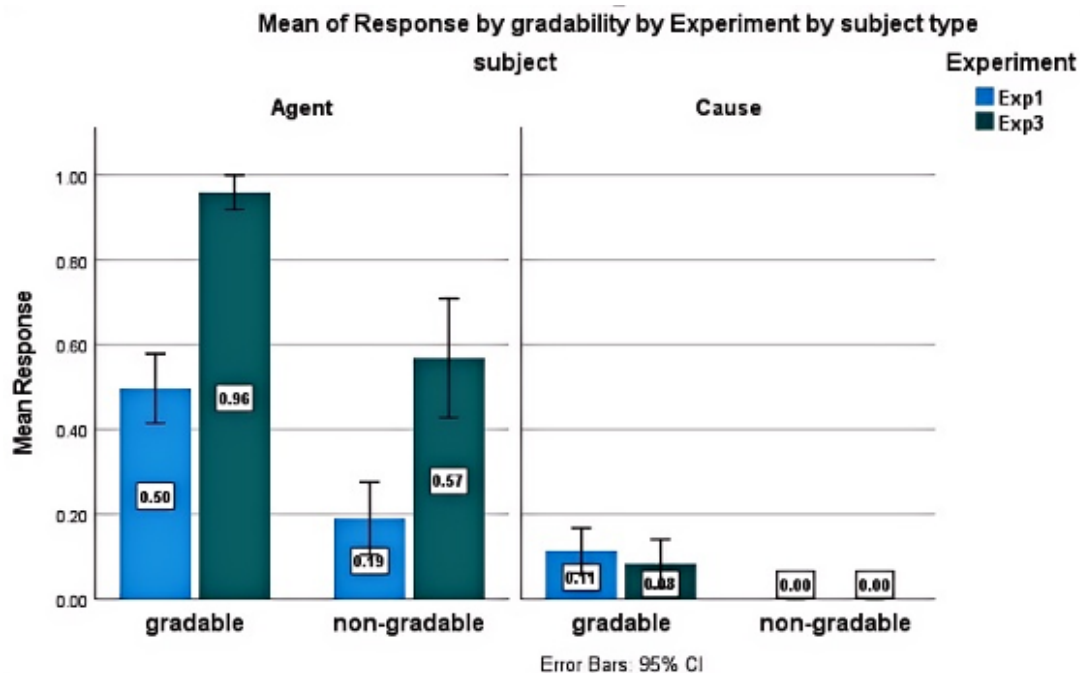


Figure 1: Results of Liu’s (2018) Experiments 1 (adverbial absent) and 3 (adverbial present) : Percentage of YES answers for zero-change situations with gradable vs non-gradable verbs, with agent subjects (left panel) and causer subjects (right panel)

(50% vs. 19%, on average). This pattern is consistent with gradability playing a role in licensing zero-change construals, thereby providing independent support for the distinction between gradable and non-gradable change-of-state SVs in Mandarin. Adding a cardinality adverbial markedly increases acceptance across both verb types—from 50% to 96% with gradable SVs and from 19% to 57% with non-gradable SVs—a difference that is statistically significant (GLMM: $\chi^2(1) = 32.14$, $p < .001$; $B = 3.27$, $OR = 26.35$). Figure 1 further shows that with an inanimate (causer) subject, perfective sentences are consistently rejected on the ZERO-CHANGE condition, irrespective of verb type.

We next turn to the intransitive use of these verbs. As Table 1 shows, the zero-change construal is available with antiagentives but categorically unavailable with anticausatives. We begin by highlighting some key differences between anticausativization in Mandarin and English.

3.4 The anticausative alternant of change-of-state SVs

3.4.1 Anticausativization in Mandarin is agentivity-insensitive

The SVs that we analyze as genuine change-of-state SVs also alternate in Mandarin, just as change-of-state verbs do in English, as illustrated in (40).

- (40) *Nèi-běn shū shāo le.*
 that-CL book burn PFV/PERF
 ‘The book (has) burned.’

Moreover, anticausatives in Mandarin ((41a)), unlike *bei*-passives ((41b)), pattern with English anticausatives (Levin and Rappaport Hovav 1995, Schäfer 2008) and with Mandarin antiagentives (recall (15)) in disallowing modification by agent-oriented adverbials.

- (41) a. *Niú* (*gùyì de) *shā* le.
Ox intentionally kill PFV/PERF
'The ox (has) (intentionally) killed.'
- b. *Niú bèi gùyì de shā* le.
Ox BEI intentionally kill PFV/PERF
'The ox was intentionally killed.'

The *by itself* test, often used to distinguish anticausatives from passives (e.g, Chierchia 2004; Schäfer 2007; Koontz-Garboden 2009; Alexiadou et al. 2015), point to the same conclusion. As shown in (42)–(43), the use of *zìjǐ* 'by itself' is compatible with the theme-subject construal only in the absence of the passive marker *bèi*. This parallels the behavior of anticausatives in languages such as Italian (Chierchia 2004) and English (Alexiadou et al. 2015: 21 and authors cited therein).

- (42) *Mén* (*bèi) *zìjǐ kāi* le.
door BEI itself open PFV
'The door (was) opened by itself.'
- (43) *Miàntuán* (*bèi) *zìjǐ fā* le.
dough BEI self leaven PFV
'The dough (was) leavened by itself.'

There is, however, another notable typological difference between Mandarin and English type languages. In English, only change-of-state verbs that do not require an agent as external argument allow an intransitive alternant (see Levin and Rappaport Hovav 1995, Piñón 2001, Reinhart 2002, Alexiadou et al. 2015 among others). Thus although *paint* is causative (Kiparsky 1997), it does not alternate because it requires an agentive external argument.

Mandarin provides two counter-examples to this generalization, corresponding to two distinct senses in which a verb may be agentive.

First, TS-oriented activity verbs such as *xǐ* 'wash' or *xiū* 'repair' are agentive in that their external argument must be an agent—that is, an individual-denoting participant in the vP-event, whether animate or inanimate—rather than a causer, namely, an eventuality-, fact-, or state-of-affairs-denoting argument that causes the vP-event. Yet, as we have seen, these verbs nonetheless alternate, yielding antiagentives.

Second, change-of-state verbs such as *shā* 'kill' are also agentive (Lin and Liu 2005: 14, Tai and Chou 1975: 50, Zhang 2025: 604), but in a different sense. They likewise alternate, yielding anticausatives, as illustrated in (41a). Unlike TS-oriented activity verbs, *shā* does not require an individual-denoting agent in subject position. Rather, it may take a causer subject denoting an event, provided that the event be construed as an action. This is illustrated in (44a)–(44b), where the subjects are action descriptions.

- (44) a. *Zǐdàn de sāoshè shā* le *Lāsīpǔjīng*.
bullet DE continuous.firing kill PFV Rasputin
'The continuous firing of the bullets killed Rasputin.'

- b. *Shāchóngjì de shǐyòng shā le nèixiē zhāngláng.*
 pesticide DE use kill PFV those cockroach
 ‘The use of the pesticide killed those cockroaches.’

This distinguishes *shā* from verbs of creation and consumption, as well as from English *murder* (and its Mandarin counterpart *móu-shā* ‘plot-kill’). English *murder* or *paint*, for instance, require an individual-denoting DP in subject position and therefore exclude event-denoting subjects. This is illustrated in (45).

- (45) #The explosion painted the workers red. (Kiparsky 1997)

The Mandarin facts therefore show that the standard binary distinction between verbs that require an agentive external argument and those that do not is insufficient. Capturing the cross-linguistic alternation patterns requires distinguishing three types of verbs: change-of-state verbs like English *paint* or *murder* that require their subject to denote an animate agent, activity verbs that require their subject to denote an agent (whether animate or inanimate), and change-of-state verbs like Mandarin *shā* that may take as subject, either an individual-denoting agent or an event-denoting causer describing an action.¹⁰

3.4.2 The zero-change reading in anticausatives

Unlike antiagentives, which license zero-change construals (cf. Section 3.2), anticausative SVs are incompatible with zero-change scenarios, as illustrated in (47)–(48).

- (47) *Mén guān le, (#dàn gēnběn méi guān-shàng).*
 door close PFV/PERF but at.all NEG.PFV close-up
 Intended: ‘The door (has) closed, but it didn’t get closed at all.’
- (48) *Shū shāo le, (#dàn gēnběn méi shāo-zháo).*
 book burn PFV/PERF but at.all NEG.PFV burn-ignite
 Intended: ‘The book (has) burned, but it didn’t get burned at all.’

The unavailability of zero-change construals with anticausatives patterns exactly with what we observed for the transitive use of the same verbs with inanimate causer subjects ((35)–(36)).

The same conclusion extends to non-gradable change-of-state SVs, such as *shā* ‘kill’, *zhāi* ‘pick’, *chú* ‘get rid of’ and *guān nài-ge shūdiàn* ‘close the bookstore’. Cardinality adverbials rescue the zero-change construal in the transitive use of these verbs, but not in their anticausative use. Thus, (49) and (50) can only receive an anomalous reading where, respectively, the ox came to life again after dying, and the apple returns to its original location after having been picked. By contrast, the corresponding transitive agentive variants are felicitous descriptions of unsuccessful attempts.

¹⁰Subjects denoting events or circumstances that are not readily construed as actions are considerably less acceptable with *shā*. Thus, (46) is at best marginal, unless the causing event is personified, or understood metonymically as standing for the human actions or policies that lead to death:

- (46) #*Zhanzheng/ na-chang baofengyu shā le Lisi.*
 war that-CL storm kill PFV Lisi
 ‘The war/that storm killed Lisi.’

(Lin and Liu 2005: 15)

We take this contrast to reflect a root-specific lexical restriction associated with *shā*: the causing event must be construed as either agentive or action-like. Within a Distributed Morphology framework, such restrictions belong to the encyclopedic content associated with individual roots (Harley 2014).

(49) #*Niú shā le yíci.*

ox kill PFV one.time

Intended: ‘The ox killed once.’

(50) #*Nèi-gè píngguǒ zhāi le hǎojiǐci.*

that-CL apple pick PFV several.times

Intended: ‘The apple picked several times.’

To conclude, the distribution of zero-change construals shows that Mandarin change-of-state SVs entail a change toward a lexically encoded state whenever no agent is projected in subject position. The intransitive result entailments of *kill* and *burn* further support treating them as distinct from TS-oriented activity SVs.

4 Deriving the zero-change construal

To account for the different licensing conditions on the zero-change construal, we propose a new semantics for Mandarin change-of-state SVs, and for gradable change-of-state SVs in particular, building on the analysis of alternating verbs developed in [Martin et al. \(2025\)](#). We spell out the proposal in Section 4.1 and provide supporting Mandarin data in Section 4.1.2.

4.1 A new semantics for Mandarin change-of-state verbs

[Martin et al. \(2025\)](#) take as their point of departure two competing analyses of change-of-state verbs. [Alexiadou et al. \(2015\)](#) decompose change-of-state verbs via a (v)-CAUSE head: in both causative and anticausative constructions, the verbal predicate denotes a set of events e that stand in a causing relation to a result state s satisfying the state predicate P encoded by the verb. [Rapp and von Stechow \(1999\)](#) likewise posit a uniform relation, but identify it as BECOME rather than CAUSE: the event e is uniformly a change developing into the result state s . Despite their differences, both analyses assume that change-of-state verbs have a single event argument and that the root, or the vP, fixes the relation between this event and the existentially quantified result state across all syntactic uses of the verb. [Martin et al. \(2025\)](#) challenge this assumption. Drawing on evidence from temporal adverbials, the progressive, and aspectual verbs, they propose that the relation between the verbal event e and the result state s remains underspecified at the vP level and is determined only in a specific syntactic context. Formally, the verbalizing head v introduces a free relational argument R , rather than a fixed causal or inchoative relation:

$$(51) \quad v_{cos} \rightsquigarrow \lambda P \lambda R \lambda e. \exists s (R(e, s) \wedge P(s))$$

The relation R is saturated when the vP is embedded in a larger syntactic structure. We now illustrate the workings of the system step by step, using *shā* ‘kill’ as our running example.

4.1.1 Extending Martin et al. (2025) to Mandarin

The core of the proposal is that Agent Voice fixes R as cause^+ , while Causer Voice—which introduces causer (event- or situation-denoting) subjects—as well as the absence of any Voice projection fixes R as become .

Since the cause^+ relation is central to our analysis, we begin by specifying its semantics. As defined in (52), whenever cause^+ relates an event e to a state s satisfying the predicate P and whose theme is an individual y , the event e decomposes into two subevents: an agentive action e' , whose

agent is x , and a change-of-state event e'' , which stands in the become relation to the result state s . The overall event e is the mereological sum of these two subevents.

$$(52) \quad \forall e \forall s \forall P \forall y (\text{cause}^+(e, s) \wedge P(s) \wedge \text{theme}(s, y) \rightarrow \\ \exists e' \exists e'' \exists x (\text{agent}(e', x) \wedge \text{become}(e'', s) \wedge e = e' \oplus e''))$$

Thus under Agent Voice, the causing event necessarily comprises both the agent's action e' and the change-of-state e'' it brings about. Because the action may begin before the change itself, the event e may have initial parts that contain no change. As we show below, this property derives the zero-change construal and distinguishes Agent Voice from all other syntactic configurations.

We now illustrate the proposal with *shā* 'kill' (here with the object *nèi-zhī zhāngláng* 'that cockroach'). At the vP level, the value of the event-state relation (R) has not yet been determined:

$$(53) \quad \llbracket \text{shā nèi-zhī zhāngláng} \rrbracket \rightsquigarrow \lambda R \lambda e. \exists s (R(e, s) \wedge \mathbf{dead}(s) \wedge \mathbf{theme}(s, \text{that-cockroach}))$$

Merged with Agent Voice (licensing *Lùlu* as the external argument), R is resolved as cause^+ . By (52), the resulting event is the mereological sum of *Lùlu*'s action and the change undergone by the cockroach:

$$(54) \quad \llbracket \text{Lulu Voice}_{ag} \text{ shā nèi-zhī zhāngláng} \rrbracket \rightsquigarrow \\ \lambda e. \mathbf{agent}(e, \text{lùlu}) \wedge \exists s (\mathbf{cause}^+(e, s) \wedge \mathbf{dead}(s) \wedge \mathbf{theme}(s, \text{that-cockroach}))$$

Merged with Causer Voice (licensing an action description such as *shāchóngyào de shǐyòng* 'the use of the insecticide'), R is resolved as become. The causer subject denotes a situation, bearing the CAUSE relation—not CAUSE^+ —to the event denoted by the verb; unlike CAUSE^+ , CAUSE relates a situation to an event.

In other words, the situation introduced by Causer Voice remains *external* to the verbal event. Rather than being identified with the verbal event (contra [Pylkkänen 2008](#), [Alexiadou et al. 2015](#)), it is simply related to it by cause. The verbal event itself is merely a change.

$$(55) \quad \llbracket \text{shāchóngyào Voice}_c \text{ shā nèi-zhī zhāngláng} \rrbracket \rightsquigarrow \\ \lambda e. \mathbf{cause}(\mathbf{the-use-of-the-insecticide}, e) \wedge \\ \exists s (\mathbf{become}(e, s) \wedge \mathbf{dead}(s) \wedge \mathbf{theme}(s, \text{that-cockroach}))$$

An important consequence of this analysis is that the same action may be semantically represented in two different ways, depending on the nature of the subject. Under Agent Voice, the subject denotes an individual agent, so the corresponding action forms part of the verbal event (as the e' component of a CAUSE^+ event). Under Causer Voice, by contrast, the subject can denote the action itself, as in action descriptions such as *shāchóngyào de shǐyòng* 'the use of the insecticide', so that the action remains external to the verbal event and merely causes it.

In the absence of any Voice projection, as in the anticausative use of *shā*, R is likewise resolved as become: the verbal event e is simply a change developing into a result state s , with no causing situation represented in the semantics (contra [Alexiadou et al. 2015](#)):

$$(56) \quad \llbracket \text{nèi-zhī zhāngláng shā le} \rrbracket \rightsquigarrow \\ \lambda e. \exists s (\mathbf{become}(e, s) \wedge \mathbf{dead}(s) \wedge \mathbf{theme}(s, \text{that-cockroach}))$$

The analysis therefore predicts a semantic split between Agent Voice, on the one hand, and Causer Voice and the absence of Voice, on the other. We now show that Mandarin exhibits precisely this Voice-conditioned split by applying two diagnostics introduced by [Martin et al. \(2025\)](#) for English and French: separate temporal modification and the progressive.

4.1.2 Evidence from Mandarin

Separate temporal modification. Martin et al. (2025) observe that when a change-of-state verb takes a causer subject, the event denoted by the subject and the verbal event may occur at non-overlapping times. This supports the view that the verbal event is merely the change, distinct from—and caused by—the event or situation denoted by the causer subject; see (57a-b).

- (57) a. The continued snowmelt **on Sunday** eventually flooded the valley **on Thursday**.
 b. Jean-Paul fell **on Sunday** from the third floor. After much agony and medical attempts to save him, the fall ended up killing him **on Tuesday night**.

The same judgments obtain in Mandarin with *yān* ‘flood’. With a causer subject (*the continued snowmelt*), the event denoted by the subject and the verbal event can be temporally separated, just as in English:

- (58) *Zhōumò* *chíxùbúduàn de róng-xuě zuìzhōng zài zhōusì yān le nèi-ge dǎi*
 Weekend continuous DE melt-snow eventually at Thursday flood PFV that-CL low
de cūnzhuāng.
 DE village
 ‘The continued snowmelt during the weekend eventually flooded that low-slung village on Thursday.’

Similarly, in (59), the subject of the second clause is the causer *dúyào* ‘the poison’, denoting the poisoning that took place on Monday; while the killing event proper (the change to being dead) occurs several days later:

- (59) *Lìsì zhōuyī gěi tā de lǎobǎn xià le dú, xiǎngyào shā-sǐ tā. Jīngguò yīshēng*
 Lisi Monday to 3SG DE boss put PFV poison want kill 3SG after doctor
duōcì qiǎngjiù, dúyào zuìzhōng háishì zài jǐtiān hòu shā-sǐ le tā.
 many.times rescue poison eventually still at several.days after kill-die PFV 3SG
 ‘Lisi poisoned her boss with poison on Monday, wanting to kill him. After several attempts by the doctors to save him, the poison eventually killed him several days later.’

The sentence is felicitous, exactly as predicted if, under a causer subject, the event denoted by *shā-sǐ* ‘kill dead’ is merely the change to being dead, the subject denoting the causing situation (cf. the semantic representation in (55)). By contrast, if the verbal event were a full cause⁺ event, as under Agent Voice (cf. the representation in (54)), it would be the mereological sum of Lisi’s action and the change undergone by his boss. Precisely because these are parts of one and the same event, they could not be modified by temporal adverbials referring to non-overlapping intervals. The sentence would therefore be contradictory. This is indeed what we find when the subject is an agent (see Fodor 1970 on English):

- (60) *Lìsì zhōuyī gěi tā lǎobǎn xià le dú. #Zuìzhōng tā (zài jǐtiān hòu)*
 Lisi Monday to 3SG boss put PFV poison eventually 3SG at several.days after
shā-sǐ le lǎobǎn.
 kill-die PFV boss
 Intended: ‘By poisoning his boss on Monday, Lisi killed him several days later.’

The progressive. The second diagnostic concerns the progressive. [Martin et al. \(2025\)](#) show that, with an agent subject, a progressive change-of-state sentence may be true *before* the change has begun, because the progressive can return an agentive action component *e'* of the cause⁺ event. By contrast, with a causer subject, the change itself must already have begun. Mandarin displays the same contrast with the progressive marker *zài*. With an agent subject, denying that any change has yet occurred is felicitous:

- (61) a. *Báisùzhēn zài yān Jīnshān-sì, dàn sì lǐ hái méi jìn-shuǐ.*
 Baisuzhen PROG flood Jinshan-temple but temple inside still NEG.PFV enter-water
 'Baisuzhen is flooding Jinshan Temple, but the water hasn't entered the temple yet.'
- b. *Tāmen zhèngzài shāo nèi duī mùtòu, dàn mùtòu gēnběn méi shāo-zháo.*
 3PL PROG burn that pile wood but wood at.all NEG.PFV burn-ignite
 'They are burning that pile of wood, but the wood didn't catch fire at all.'

But with a causer subject, the progressive sentence implies that the change has already begun, rendering the same continuation infelicitous.

- (62) a. *Hóngshuǐ zài yān Jīnshān-sì, #dàn sì lǐ hái méi jìn-shuǐ.*
 flood.water PROG flood Jinshan-temple but temple inside still NEG.PFV enter-water
 Intended: 'The floodwater is flooding Jinshan Temple, but the water hasn't entered the temple yet.'
- b. *Shān-huǒ zhèngzài shāo nèi duī mùtòu, #dàn mùtòu gēnběn méi shāo-zháo.*
 mountain-fire PROG burn that pile wood but wood at.all NEG.PFV burn-ignite
 Intended: 'Wild fire is burning that pile of wood, but the wood didn't catch fire at all.'

The Mandarin data thus replicate the English and French pattern reported by [Martin et al. \(2025\)](#): the verbal event described by a change-of-state verb is understood as beginning only when the change begins, unless the subject is an agent. This follows directly from the analysis. Except under Agent Voice, the causing situation remains external to the verbal event, which is therefore identified with the change itself and lacks a change-free initial part. The zero-change construal is accordingly predicted to arise only under Agent Voice.

The analysis developed in this section explains why the zero-change construal is restricted to Agent Voice configurations. What remains to be explained is which semantic ingredients license this construal.

4.2 Two sources for the zero-change construal of change-of-state SVs

We propose that the zero-change construal of Mandarin change-of-state SVs may be licensed by two potentially cumulative sources: the degree argument available with gradable verbs and verbal *le*, which we analyze as a non-completive perfective marker. When both sources are active, the construal is more readily available than when only one is. This explains why the construal is accepted more readily with gradable than with non-gradable change-of-state SVs: gradable verbs draw on both sources, whereas non-gradable verbs rely only on verbal *le* (recall Liu's 2018 experiments, which found acceptance rates of 50% vs. 19% in the absence of a cardinality adverbial, and 96% vs. 56% in its presence).

A revealing, and to our knowledge previously unnoticed, contrast between gradable vs. non-gradable change-of-state SVs is illustrated in (63)–(66), all of which involve a bare change-of-state verb (i.e. WITHOUT aspectual marking) in combination with a cardinality adverbial. Gradable change-of-state SVs combine felicitously with such adverbials, whereas non-gradable ones do not. Thus, (63) can be used to instruct someone to try at least twice to burn a book that is difficult to burn. By contrast, Yusupov (the man believed to have orchestrated Rasputin’s assassination) could not felicitously use (64) to instruct his co-conspirators to try several times to kill Rasputin because he had proven difficult to kill. As in English, the sentence instead suggests anomalously that Rasputin could be successfully killed several times.

- (63) *Wǒ mìnglìng nǐ shāo nà-běn shū liǎng-cì.*
 1SG order 2SG burn that-CL book two-time
 ‘I order you to burn that book twice.’
- (64) #*Wǒ mìnglìng nǐ shā Lāsīpǔjīng sān-cì.*
 1SG order 2SG kill Rasputin three-time
 Intended: ‘I order you to kill Rasputin three times.’
- (65) *Shāo nà-běn shū liǎng-cì shì ge hǎo zhúyì.*
 burn that-CL book two-time be CL good idea
 ‘To burn that book twice is a good idea.’
- (66) #*Shā Lāsīpǔjīng sān-cì shì ge hǎo zhúyì.*
 kill Rasputin three-time be CL good idea
 Intended: ‘To kill Rasputin three times is a good idea.’

This supports our claim that the two subclasses differ in the locus of non-culmination. We examine the first source in Section 4.3 and the second in Section 4.4.

4.3 First source: the degree argument

4.3.1 Degree of event realization vs. degree of change

As mentioned in the introduction, our treatment of gradable change-of-state verbs is inspired by Piñón’s (2008) analysis of incremental theme and degree achievement verbs. Central to Piñón’s proposal is the notion of an event type’s degree of realization. To capture this notion, he assumes that incremental theme and degree achievement verbs are lexically associated with measure functions mapping individuals x , descriptions O , and events e to degrees:

- (67) a. **eat** _{δ} (x)(O)(e) ‘the degree to which x qua type O is eaten in e ’
 b. **read** _{δ} (x)(O)(e) ‘the degree to which x qua type O is read in e ’
 c. **lengthen** _{δ} (x)(O)(e) ‘the degree to which x qua type O is lengthened in e ’

The corresponding verbs then denote relations between individuals x , descriptions O , degrees d , and events e :

- (68) **eat** $\rightsquigarrow \lambda x \lambda O \lambda d \lambda e. \text{eat}_\delta(x)(O)(e) = d$

The degree argument can be discharged in either of two ways. The POSITIVE BINDING OPERATOR (henceforth POS) restricts its value to be greater than zero, whereas the DEGREE MAXIMIZING OPERATOR (henceforth MAX) sets the value of the degree argument to 1. While we adopt a variant of Piñón’s (2008) proposal, we depart from it on one point. For Piñón, the measure function as-

sociated with a verb returns the degree to which x qua type O is AFFECTED (or effected) in e with respect to the relevant property.

Such a definition, however, presupposes that the theme is always affected to some positive degree. While this assumption is harmless for English, it becomes problematic when extended to languages such as Mandarin, whose change-of-state verbs admit construals where the theme need not undergo any change whatsoever.

We therefore take Piñón's (2008) notion of the degree of realization of an event type more literally. We assume that the measure function merely measures the extent to which the relevant event type is realized, giving up the assumption that a positive degree of event realization entails a positive degree of change. On this proposal, the degree of event realization may be positive even when the theme has not (yet) been affected at all.

Thus, for instance, the measure function burn_δ which underlies the meaning of *shāo shū* 'burn the book' determines the degree d_e to which a burning-the-book event is realized in e .

The representation of *shāo* at the vP level (with the object *shū* 'the book' merged) is given in (69). Compare this with the corresponding representation of non-gradable *shā* in (53).

$$(69) \quad \llbracket \text{shāo shū} \rrbracket \rightsquigarrow \\ \lambda R \lambda d_e \lambda e. \exists s (R(e, s) \wedge \text{burnt}(s) \wedge \text{theme}(s, \text{the-book}) \wedge \text{burn}_\delta(e, \text{the-book}) = d_e)$$

When $d_e = 1$, the event has reached its maximal degree of realization, guaranteeing that there is a result state s in which the book is burnt. Since the adverb of completion *quán* 'completely' in (28) requires maximal event realization, the zero-change continuation is infelicitous.

4.3.2 The interaction between the degree argument and Agent Voice

We are now in a position to explain why gradable change-of-state SVs used agentively license the zero-change construal, even in the absence of outer aspect morphology.

Merged with Agent Voice, R is resolved as cause^+ , and Mòmo is introduced as the agent, yielding (70):

$$(70) \quad \llbracket \text{Mòmo Voice}_{ag} \text{ shāo shū} \rrbracket \rightsquigarrow \\ \lambda d_e \lambda e. \text{agent}(e, \text{mòmo}) \wedge \\ \exists s (\text{cause}^+(e, s) \wedge \text{burnt}(s) \wedge \text{theme}(s, \text{the-book}) \wedge \text{burn}_\delta(e, \text{the-book}) = d_e)$$

The key lies in the interaction between the degree argument d_e and Agent Voice. By (52), cause^+ decomposes the verbal event e into the mereological sum of the agent's action e' and the change-of-state e'' : $e = e' \oplus e''$. Because the action may begin before the change, the event can be realized to a positive degree ($d_e > 0$), while the theme has undergone no change at all ($d_c = 0$). This is the case, for instance, in (70), if Mòmo has prepared the fire and placed the book into it, but the book is too damp to have started burning. The event is therefore realized to a positive degree, even though the change itself has not yet begun.

The degree of event realization d_e thus tracks event progress, rather than change progress. Accordingly, degree modifiers such as *shāowēi* 'slightly' or *yíxiǎ* 'a little' (which may also co-occur) require d_e to take a positive but low value, thereby blocking MAX. Since an agentive causation event can be realized to a positive degree without the theme undergoing any change at all (i.e. $d_e > 0$, while $d_c = 0$; d_c being the degree of change), the zero-change construal remains available even in the presence of these modifiers, as shown in (71):

- (71) *Nèi-běn shū Mòmo zhǐ shāowēi shāo le yíxià, dàn gēnběn méi shāo-zháo.*
 that-CL book Momo only slightly burn PFV a.little but at.all NEG.PFV burn-ignite
 ‘Momo only slightly burned the book, but it didn’t burn at all.’

That the continuation in (71) denies any change whatsoever confirms that Mandarin gradable change-of-state SVs encode a degree of event realization, rather than a degree of change. If the degree argument measured change, as in Piñón’s original proposal, any positive value of the degree argument would entail some degree of change, making the zero-change continuation contradictory in (71).¹¹

By contrast, in the absence of Agent Voice, a positive value of d_e necessarily implies that the theme has undergone some degree of change. As shown in Section 4.1, without an agent, the verbal event is nothing more than the change itself. Consequently, realizing the event to a positive degree amounts to realizing the change to a positive degree.

Finally, recall that Liu’s (2018) findings show that, in the absence of a cardinality adverbial, the zero-change construal of gradable change-of-state SVs is accepted only about half of the time (50% compared with 96% when a cardinality adverbial is present). We attribute this lower acceptance rate to a preference for interpreting the degree argument d_e bound by MAX rather than by POS. Following Piñón (2008), we assume that MAX is the default binder, since it yields a stronger interpretation: $d_e = 1$ entails $d_e > 0$, but not conversely (see also Kennedy and Levin 2008 on the preference for telic interpretations of optionally telic predicates). As a result, a perfective sentence such as *Lǐsì shāo le yí-gēn mùtóu* ‘Lisi burned a piece of wood’ is by default interpreted as describing an event realized to its maximal degree, which guarantees the existence of a result state (here, that the piece of wood was burnt). Deriving the zero-change construal therefore requires overriding this preferred interpretation. Participants who reject the zero-change scenario are precisely those who do not override this default preference.

4.3.3 The cardinality adverbial effect with gradable change-of-state SVs

We now turn to the question of why cardinality adverbials facilitate the zero-change reading with gradable change-of-state SVs. Under de Swart’s (1991) PLURALITY CONDITION, a cardinality adverbial such as *hǎojiǐcì* ‘several times’ cannot felicitously quantify over a set known to contain fewer than two events. Consequently, it is incompatible with ONCE-ONLY PREDICATES, as illustrated by #*Lisi killed the ox twice* in English. Following de Swart (1991: 65), a predicate is once-only if it carries a UNIQUENESS PRESUPPOSITION on its Davidsonian event argument, so that for every model and every assignment of individuals to its arguments, it denotes a singleton set of events. Thus, for a fixed agent and theme, *kill the ox* denotes a single complete killing event (Lisi can kill the same ox only once).¹²

If the degree d_e encoded by the gradable change-of-state VP *shāo shū* ‘burn the book’ is set to 1, the VP is a once-only predicate, since the same book can be fully burned only once. By contrast, if $1 > d_e > 0$, the VP denotes a set of events whose cardinality can exceed one: there can be many failed attempts to burn the same book. A cardinality adverbial therefore forces POS as the binder of d_e , overriding the otherwise preferred MAX.

In other words, cardinality adverbials serve as an explicit cue to discard the stronger meaning,

¹¹We thank [removed for review] for discussion on this point.

¹²Since this requirement is relativized to a model and an assignment, it is not falsified by the fact that, across models, the same individuals may be related by the predicate in different ways (e.g. through different killing events).

thereby favouring the positive-degree reading under which the zero-change construal is available.

English *burn* appears to pattern with Mandarin *shāo* in allowing cardinality adverbials, as shown in (72). Here too, the VP *burn the dress* denotes a set of events whose cardinality exceeds one. The similarity with Mandarin is only apparent, however. Even under the non-maximal interpretation forced by the cardinality adverbial—where the definite is understood as referring to different subparts of the dress (Križ, 2016)—each event still entails some change in the relevant subpart of the theme. The zero-change continuation is therefore contradictory.

- (72) Damn it! How many times have I burned my dress while ironing it! #Fortunately, it is still completely intact!

We capture this cross-linguistic contrast by assuming that *burn* and Mandarin *shāo*, though both gradable, encode different kinds of degree. As argued above, Mandarin *shāo* encodes a degree of *event realization* d_e , whereas English *burn* encodes a *change* d_c in the sense of Kennedy and Levin (2008). Since d_c directly measures the extent to which the theme has acquired the relevant property, any positive value of d_c entails that some change has occurred. Consequently, whether d_c is bound by POS or by MAX, the VP denotes only events in which the theme has undergone some degree of change, making the zero-change continuation impossible.

To sum up, the degree argument provides one source of the zero-change construal, available only with gradable change-of-state SVs. The zero-change construal of non-gradable verbs such as *shā* ‘kill’ must therefore arise from a different source, to which we now turn.

4.4 Second source: verbal *le*

The degree argument cannot account for the zero-change construal of non-gradable change-of-state SVs, since these verbs lack such an argument. To identify its source, we compare the compatibility of gradable and non-gradable predicates with cardinality adverbials, both in their bare and perfectivized uses. We repeat the relevant contrasts in (73).

- (73) a. Wǒ mìnglìng nǐ shāo nà-běn shū liǎng-cì.
1SG order 2SG burn that-CL book two-time
‘I order you to burn that book twice.’
b. #Wǒ mìnglìng nǐ shā Lāsīpǔjīng sān-cì.
1SG order 2SG kill Rasputin three-time
Intended: ‘I order you to kill Rasputin three times.’
c. Gōngjué shā le Lāsīpǔjīng hǎojǐcì, Lāsīpǔjīng dōu méi sǐ.
prince kill PFV Rasputin several.time Rasputin DOU NEG.PFV die
Literally: ‘The prince killed Rasputin several times, but Rasputin didn’t die.’

The contrasts in (73a) and (73b) bring out an important difference between gradable and non-gradable change-of-state VPs. Whereas the bare (uninflected) VP *shāo nà-běn shū* ‘burn that book’ is not necessarily a once-only predicate—it becomes one only when $d_e = 1$ —the bare VP *shā Lāsīpǔjīng* ‘kill Rasputin’ is. Without a degree argument, the VP is necessarily a once-only predicate: for a fixed agent and theme, each model contains only one complete killing of Rasputin.

Why, then, can *kill*-predicates combine with cardinality adverbs once perfectivized, as in (73b), but not when bare, as in (73)? In other words, why are they once-only predicates when bare, but not when perfectivized? We show below that, under our analysis of *le* as a partitive, non-

completive perfective marker, a perfectivized predicate may denote several maximal parts of possible (complete) killing events—for instance, several unsuccessful attempts to kill the same individual. Before spelling out the formal analysis, we provide some background on the semantics of *le*.

4.4.1 The semantics of verbal *le*

Verbal *le* in Mandarin is generally analyzed as a perfective marker (Lin 2006, Sun 2014 among others). Under the standard neo-Reichenbachian analysis, perfective aspect relates the temporal trace of the eventuality, $\tau(e)$, to the topic time t_T , such that $\tau(e) \subseteq t_T$ (Kratzer 1996), while introducing existential closure over the event. This captures the intuition that a perfective sentence depicts an eventuality in its entirety: the event is viewed as complete.¹³

It is well established, however, that Mandarin *le* does not systematically entail event completion. Thus, Soh and Gao (2007: 349) argue that in accomplishment sentences with verbal *-le*, “the event can be terminated without having reached the inherent endpoint”. Similarly, Smith (1997: 271) argues that Mandarin *le* does not always give rise to the *closed* reading associated with perfective viewpoints in other languages. On the basis of Thai, Koenig and Muansuwan (2000) distinguish two types of perfective operators. Martin and Gyarmathy (2019) refer to these as STRONG and WEAK perfectives. STRONG perfectives entail event completion when combined with telic predicates. WEAK perfectives, by contrast, require only event maximality (Altshuler 2014): the reported event must have ceased—that is, it must be the maximal part of some possible *P*-event in the context—but it need not have culminated. Following Martin and Gyarmathy, we argue that Mandarin verbal *le* is a weak perfective.

Our analysis extends these proposals in two respects. First, we show how the non-completive semantics of *le* interacts with the event structure of the predicate to explain the cardinality adverbial effect. Second, we integrate the non-completive semantics of *le* into the framework of Martin et al. (2025). Roughly, under Agent Voice, the weak perfective may return action-only maximal parts of possible causation events, thereby deriving the zero-change construal. We now spell this out formally.

4.4.2 The interaction between *le* and Agent Voice

We attribute to *le* the meaning in (74), following the line of proposals just reviewed.¹⁴

¹³Many authors distinguish verbal *le*, analyzed as a perfective marker, from sentential *le*, analyzed as an inchoative marker (Li and Thompson 1981: 238, Paul 2015), or as a perfect marker. All the transitive examples discussed here involve verbal *le*. Some authors (e.g. Sybesma 1997, Lin 2003) have argued that verbal *le* also licenses ongoing-event readings. We do not find the available evidence is conclusive. Sybesma’s examples involve a sentence final *le* and are therefore compatible with an analysis involving sentential *le* (see Soh 2009). ?’s (?)266–267]Lin03 uncontroversially verbal examples have been reanalyzed by Soh and Gao (2007: 358–359), who argue that the event itself has ceased, while only the resulting state is (defeasibly) implied to hold at utterance time. Furthermore, although sentences with activity predicates and verbal *le* more readily allow ongoing continuations, Martin and Gyarmathy (2019) show that this pattern is also found with perfective markers in languages such as French and Hindi. They attribute this greater tolerance for ongoing readings to the way activity events are individuated. Taken together, these observations support the view that verbal *le* entails event cessation, not completion, an assumption we adopt here.

¹⁴Since we argue that *le* composes with the predicate before combining with the cardinality adverbial, we separate existential closure over events from its meaning, following Matthewson’s (2012) analysis of Gitksan).

- (74) $\llbracket \text{PFV}_M \rrbracket = \lambda P \lambda e [\tau(e) \subseteq t_T \wedge \text{MAX}(e, P)]$
 ('MAX' stands for maximality)¹⁵

Applied to the agentive predicate *Yóusūbōfū Voice_{ag} shā Lāsīpǔjīng* 'Yusupov kill Rasputin', the weak perfective returns events that are maximal parts of possible killing events. Accordingly, the events it returns need not themselves be complete killing events:

- (75) a. $\llbracket \text{Yóusūbōfū Voice}_{ag} \text{ shā Lāsīpǔjīng} \rrbracket \rightsquigarrow$
 $\lambda e. \text{agent}(e, \text{yusupov}) \wedge \exists s (\text{cause}^+(e, s) \wedge \text{dead}(s) \wedge \text{theme}(s, \text{rasputin}))$
 b. $\llbracket \text{PFV}_M \text{ Yóusūbōfū Voice}_{ag} \text{ shā Lāsīpǔjīng} \rrbracket \rightsquigarrow$
 $\lambda e [\tau(e) \subseteq t_T \wedge \text{MAX}(e, \lambda e. \text{agent}(e, \text{yusupov}) \wedge$
 $\exists s (\text{cause}^+(e, s) \wedge \text{dead}(s) \wedge \text{theme}(s, \text{rasputin})))]$

Under Agent Voice, the verbal event is a cause^+ event—that is, the mereological sum of Yusupov's action e' and Rasputin's change of state e'' (by (52)). Consequently, the events returned by the weak perfective need not include the change-of-state component. They may instead consist solely of the action e' , yet still qualify as maximal parts of possible killing events. In the appropriate context, such events correspond to unsuccessful attempts to kill Rasputin. This derives the zero-change reading in the agentive variant of these verbs.

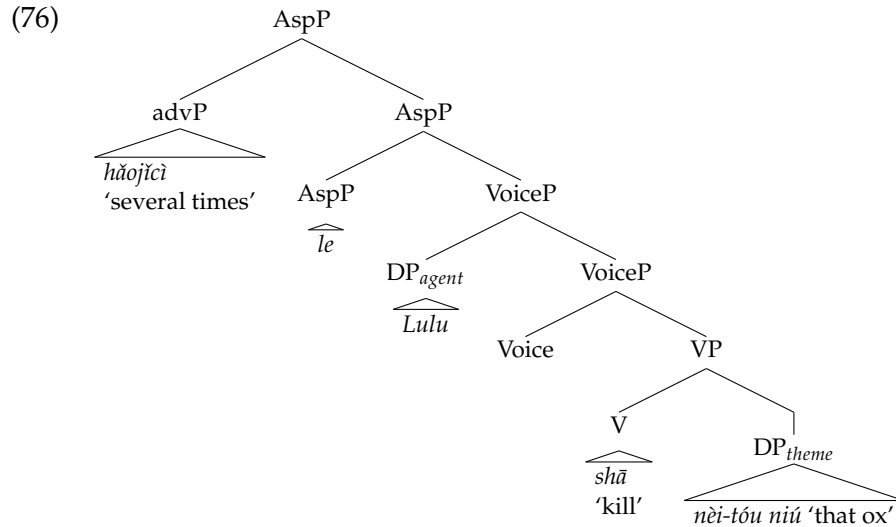
By contrast, when perfective *le* applies to the anticausative predicate ((56)); see (47b)–(48b) and (49), or to a causative predicate with a causer subject ((55); see (35)–(36)), the events returned by the weak perfective necessarily contain a change of state, because the predicate denotes only BECOME events, with no action component. Asserting in the subsequent discourse that the theme has undergone no change whatsoever is therefore infelicitous: even a proper part of a change is still a change.

Why, then, is the acceptability of the zero-change construal of perfectivized non-gradable change-of-state SVs so low in the absence of a cardinality adverb (19% of acceptance in Liu's 2018 Experiment 1; recall (33a))? We take this low acceptability to show that, although partitive perfectives such as Mandarin *le* license both completive and non-completive construals, the completive construal is preferred by default. This preference arises because the completive construal yields the stronger meaning: it entails the non-completive construal, but not conversely (intuitively, if there exists a complete P -event, then there necessarily exists a maximal part of such an event, but not conversely). Participants who reject the zero-change scenario retain this preferred, stronger of *le*, rather than shifting to the weaker non-completive construal required to make the discourse coherent.

4.4.3 The cardinality adverbial effect with non-gradable change-of-state SVs

The remaining question is why the zero-change construal of perfectivized non gradable causative SVs becomes more acceptable in the presence of a cardinality adverb. Our answer is that cardinality adverbials override the otherwise preferred completive construal of *le*. The analysis rests on two assumptions. First, the aspectual marker *le* can scope below cardinality (and temporal) adverbials, composing with the VP before the adverbial does. This is illustrated in (76).

¹⁵We follow Altshuler (2014) in taking $\text{MAX}(e, P)$ to hold iff (i) e is a part of a possible P -event, and (ii) e is not a proper part of any actual event that is itself part of a possible P -event. This informal characterization suffices for the present purposes; see Koenig and Muansuwan (2000) and Altshuler (2014) for detailed discussion and Martin and Gyarmathy (2019) for a comparison with alternative formulations of weak perfectivity.



Second, once *le* has applied to the bare predicate, the resulting predicate is no longer once-only: its Davidsonian argument may be instantiated by several distinct events, each a maximal part of a possible *P*-event. The uniqueness presupposition on the event argument is therefore suspended, making the predicate compatible with cardinality adverbials. Thus, the cardinality adverbial counts in (76) are maximal parts of kill-Rasputin events.

Importantly, the cardinality adverbial promotes the non-completive construal of *le* before the boundary of the first clause is reached. Bott (2010) and Bott and Hamm (2014) show that the culmination inference of ‘optionally completive’ predicates is readily canceled while the clause is still being processed, but becomes much harder to override once the clause boundary has been crossed. Thus, the denial of Rasputin’s death in the second clause in (33a) comes too late, whereas the cardinality adverbial in (33b) cues the non-completive construal before commitment to the default completive interpretation.

The same analysis explains the striking contrast in (77). The zero-change construal is licensed by the cardinality adverbial *yíci* ‘one time’, but not *yòu* ‘again’.

- (77) a. *Nóngfū shā le nèi-tóu niú yíci, niú méi sǐ.*
 Farmer kill PFV that-CL ox one.time ox NEG.PFV die
 Literally: ‘The farmer killed that ox one time, but the ox didn’t die.’
 b. #*Nóngfū yòu shā le nèi-tóu niú, (niú méi sǐ.)*
 Farmer again kill PFV that-CL ox ox NEG.PFV die
 Intended: ‘The farmer killed that ox again, but the ox didn’t die.’

Under the restitutive reading, *yòu* presupposes the ox’s death, making the assertion of a further killing event infelicitous. Under the repetitive reading, it presupposes a previous killing event. Since, in the absence of a cardinality adverbial, the sentence is by default interpreted as entailing the ox’s death, the subsequent denial is contradictory.

To sum up, gradable and non-gradable change-of-state SVs differ in the source of the zero-change construal. For gradable SVs, two sources are available: the degree argument d_e and particle perfective *le*. This predicts the gradient pattern observed by Liu (2018). Acceptance is highest with perfectivized gradable SVs modified by a cardinality adverbial (96%), where both sources are available and the adverbial overrides the preference for the stronger reading. It is interme-

diate with unmodified gradable SVs (50%), where both sources are also available but there is no cue favouring the weaker reading, and with modified non-gradable SVs (56%), where only the partitive perfective is available but the adverbial cues the non-completive reading. Finally, acceptance is lowest with unmodified non-gradable SVs (19%), where only the partitive perfective is available, and there is no facilitating effect of the adverbial.

5 The zero-change reading with completive markers

The lexical distinction between TS-oriented activity SVs and change-of-state SVs put forward here receives further independent support from a previously unnoticed diagnostic based on the zero-change construal. Specifically, the zero-change construal survives a completive marker (*completely, in ten minutes*) with TS-oriented activity SVs but not with change-of-state SVs (Section 5.1). This asymmetry, in turn, argues against an alternative modal analysis of Mandarin change-of-state SVs (Section 5.2).

5.1 Two types of endpoints

This diagnostic presupposes that both TS-oriented activity and change-of-state SVs form telic VPs. Otherwise, neither *in*-adverbials, nor completion adverbials such as *quán* ‘completely’ would be licensed, making any contrast between the two classes uninformative. *In*-adverbials apply felicitously only to telic predicates (Vendler 1957, Verkuyl 1972, Krifka 1989), measuring the interval between the event’s onset and its telos. We thus use compatibility with the latter as a diagnostic to show that, with quantized objects, both classes form telic VPs, unlike pure activity SVs.

Mandarin bare time adverbs such as *wǔ fēnzhōng* ‘five minutes’ are ambiguous between frame (*in*-) and durative (*for*-) readings, with the former preferred preverbally and the latter postverbally (Xiao and Mcenery 2006, Zhang 2016, among others). We therefore use the preverbal form *nèi* ‘within’ to unambiguously enforce the frame reading.¹⁶

- (78) *Lùlu wǔ fēnzhōng nèi jiù xǐ le yí-jiàn chènshān.* (TS-oriented activity SV)
 Lulu five minute within JIU wash PFV one-CL shirt
 ‘Lulu washed a shirt in five minutes.’
- (79) *Lùlu yì fēnzhōng nèi jiù shāo le nèi-běn shū.* (gradable change-of-state SV)
 Lulu one minute within JIU burn PFV that-CL book
 ‘Lulu burned that book in one minute.’
- (80) *Lùlu wǔ fēnzhōng nèi jiù shā le nèi-tóu niú.* (non-gradable change-of-state SV)
 Lulu five minute within JIU kill PFV that-CL ox
 ‘Lulu killed that ox in five minutes.’

With a quantized object, both TS-oriented activity SVs such as *xǐ* ‘wash’ (78) and causative SVs such as *shāo* ‘burn’ (79) or *shā* ‘kill’ (80) accept the frame adverbial, diagnosing a telic VP. By contrast, pure activity SVs such as *chuí* ‘hammer’ or *tuī* ‘push’ reject the frame adverbial (81) in a default context, even with a quantized object, confirming their inherent atelicity.

¹⁶To ensure a quantized object when testing telicity, we systematically use the demonstrative *nèi* ‘that’ or the numeral *yí* ‘one’, which blocks the bare plural interpretation of the noun.

- (81) #*Lùlu wǔ fēnzhōng nèi jiù chuí le nèi-ge pánzi.* (pure activity SV)
 Lulu five minute within JIU hammer PFV that-CL plate
 Intended: ‘Lulu hammered the plate in five minutes.’

The first two classes combine felicitously with *quán* ‘completely’, as expected if they are telic. They differ, however, in whether the zero-change reading survives the adverbial, it does with TS-oriented activity SVs (82b), but not with change-of-state SVs (82a).

- (82) a. *Wǒ-de shū Mòmo (#quán) shāo le, dànshì gēnběn méi shāo-zháo.*
 1SG-DE book Momo completely burn PFV but at.all NEG.PFV burn-ignite
 ‘Momo (completely) burned my book, but it didn’t burn at all.’
 b. *Wǒ-de yīfu Mòmo quán xǐ le, dànshì gēnběn méi xǐ-gānjìng.*
 1SG-DE clothes Momo completely wash PFV but at.all NEG.PFV wash-clean
 ‘Momo completely washed my clothes, but they didn’t at all get clean.’

The contrast is straightforward. Completion adverbials require the event to reach its endpoint. For the reported washing event, this endpoint is simply the end of the washing procedure applied to the whole theme, independently of whether it succeeds in making the theme clean. The target state is only conventionally associated with the verb and inferred abductively (Section 3.1). By contrast, change-of-state SVs such as *shāo* ‘burn’ lexically encode a result state. Their event type is maximally realized only when the causing event reaches the onset of that result state (Higginbotham 1995, 2000). A change of the theme is therefore entailed.

The same contrast arises with *in*-adverbials: the zero-change construal survives with *xǐ* ‘wash’ (83b) but not with *shāo* ‘burn’ (83a).

- (83) a. *Lùlu (#yì fēnzhōng nèi) shāo le nèi-běn shū, dàn shū gēnběn méi shāo-zháo.*
 Lulu one minute within burn PFV that-CL book but book at.all NEG.PFV burn-ignite
 ‘Lulu burned the book (in one minute), but the book didn’t burn at all.’
 b. *Lùlu wǔ fēnzhōng nèi xǐ le nèi-jiàn tīxù, dàn tīxù gēnběn méi xǐ-gānjìng.*
 Lulu five minute within wash PFV that-CL t-shirt but t-shirt at.all NEG.PFV wash-clean
 ‘Lulu washed that t-shirt in five minutes, but it didn’t get clean at all.’

The explanation is the same. Since TS-oriented activity SVs do not lexically encode a result state, the *in*-adverbial measures the interval to the endpoint of the TS-oriented activity itself. By contrast, with change-of-state SVs it measures the interval to the onset of the lexically encoded result state. A change of the theme is therefore entailed.¹⁷

¹⁷A potential concern arises at this point. If Mandarin verbal *le* is a partitive perfective (Section 4.4), why do adverbials such as *yì fēnzhōng nèi* ‘in one minute’ or *quán* ‘completely’ force a completive construal, as we have tacitly assumed? This empirical puzzle has already been noted by Soh and Gao (2007) who observe that although verbal *le* generally permits both non-completive (‘terminative’) and completive construals, adding a frame *in*-adverbial to a perfective change-of-state SV sentence forces the completive construal, and hence attainment of the result. They do not, however, explain why.

At first sight, this is puzzling. Completion adverbials do not by themselves force culmination with all partitive

To summarize, both TS-oriented activity and change-of-state SVs are telic, but their endpoints differ: for TS-oriented activities, the endpoint is that of the TS-oriented activity itself; for change-of-state SVs, it is the onset of the encoded result state. This is why the zero-change construal is ruled out with change-of-state SVs in the presence of adverbials marking event completion.

5.2 Against a modal analysis of Mandarin change-of-state SVs

The interaction between the zero-change construal and adverbials entailing event completion also bears on an alternative analysis of Mandarin change-of-state SVs. So far, we have attributed the zero-change construal to two sources: the verb's degree argument (Section 4.3) and our analysis of *le* as a weak (partitive) perfective (Section 4.4). An alternative, suggested by a reviewer, is to derive the zero-change reading without appealing to either source—that is, without assuming either a lexical difference between English and Mandarin *burn* or our partitive analysis of *le*.

On this alternative, the causative change-of-state SV lexically encodes a result state, but the latter falls within the scope of a modal operator, so that its realization is not asserted in the evaluation world. Existing proposals differ both in the locus of this operator and the modal base it quantifies over. For Beavers and Lee (2020), the operator is introduced by Agent Voice and quantifies over the external argument's intention worlds I_x . For Martin and Schäfer (2012), it is introduced within the *vP*, with the flavor of the accessibility relation ρ_w determined by the subject. For Martin (forthcoming), it is introduced by the verbalizing head, with the flavor largely determined by context, as with ordinary modal verbs. Despite these differences, the resulting truth conditions are broadly similar. Combining a Beavers and Lee-style account of Mandarin with the event semantics of Martin et al. (2025), the agentive use of *shāo* 'burn' receives the representation in (84):

$$(84) \quad \llbracket \text{Lùlu Voice}_{ag} \text{ shāo nèi-běn shū} \rrbracket \rightsquigarrow \\ \lambda d_e \lambda e \lambda w. \text{agent}(e, \text{lùlu}) \wedge \text{burn}_\delta(e, \text{the-book}) = d_e \\ \wedge \Box_{I_{\text{lùlu}}^{w'}} \exists s (\text{cause}_{w'}^+(e, s) \wedge \text{burnt}_{w'}(s) \wedge \text{theme}(s, \text{the-book})) \\ (\text{Condition: } I_{\text{lùlu}} \neq \emptyset)$$

In (84), only the agent's action e is asserted at the evaluation world w ; the change-of-state remains within the scope of the modal. The zero-change reading thus arises whenever $w \notin I_{\text{lùlu}}$. Since

aspectual operators. As Higginbotham (2004: 343) observes, *in*-adverbials, and we add *completely*, are compatible with the English progressive, an imperfective partitive operator (e.g. *Just as I arrived, John was running a mile in ten minutes; John was rewriting the manuscript completely*), even though such sentences still give rise to the imperfective paradox: the adverbial measures the interval to a telos (John's having run a mile), but the telos need not be reached (John may have been interrupted after a few hundred meters). If Mandarin *le* is likewise a partitive, non-completive operator, why should adding a completion adverbial to a *le*-sentence force a completive construal?

The answer lies in the difference between an imperfective and a partitive perfective. The progressive is an imperfective operator: the runtime of the asserted event properly includes the topic time ($\tau(e) \supset t_T$). The event is therefore viewed as still in progress, and whether its telos is ultimately reached remains open (Dowty 1979; Altshuler 2016: 111–112). Consequently, an *in*-adverbial may measure the interval to a telos that need not coincide with the right boundary of the asserted event.

Mandarin *le*, by contrast, is a partitive perfective. Although it need not denote a complete event, it returns a maximal stage of a possible *P*-event. The asserted event therefore cannot continue into a larger possible *P*-event. Consequently, the completion adverbial cannot locate the telos beyond the asserted event, but must identify it with the right boundary of that event. Event completion is therefore forced.

A full discussion would require a more detailed comparison between Mandarin *le* and the English progressive, and more generally between imperfective and partitive perfective aspect. Since this would take us beyond the main line of argument, we confine it to this footnote.

the modal operator is contributed by active Voice, this construal is predicted to be restricted to agentive uses.

The question then, is whether a modal analysis extends to Mandarin. We suggest that it does not. The relevant evidence comes from adverbials entailing event completion. Unlike Mandarin change-of-state SVs, modal causatives in English, French and German retain the zero-change construal under completion adverbials:

- (85) a. Audrey explained the problem to Moritz (in ten minutes), but he did not understand it.
(Martin and Demirdache 2020)
b. Audrey (completely) explained the problem to Moritz, but he did not understand it.

With modal causatives, the endpoint that serves as the right anchor for completion adverbials differs depending on whether the action is successful or not. If it is, the endpoint coincides with the onset of the result state, as with ordinary causative verbs. If it is not, as in (85), the endpoint is the point at which the agent's intended action is completed—for instance, when Audrey has finished explaining the problem. Only in the relevant modal continuations does this endpoint coincide with Moritz's coming to understand the problem.

If a modal analysis such as those of Martin and Schäfer (2017) or Beavers and Lee (2020) were extended to Mandarin, agentive change-of-state SVs would be expected to pattern like English defeasible causatives: on representations such as (7b) and (84), they should retain the zero-change construal when modified by completion adverbials. As Section 5 has shown, however, they do not. The Mandarin data thus favor a non-modal analysis of change-of-state SVs.

6 Antiagentivity and anticausativity as hallmarks of telicity

Cheng (1989) observed a striking generalization about the Mandarin lexicon: exactly those SVs whose transitive uses form telic predicates with a quantized object admit an intransitive alternant. Remarkably, this correlation remains exceptionless even under the finer three-way typology of Mandarin SVs argued for in this paper, which distinguishes TS-oriented activity SVs from pure activity SVs. This section explains why.

6.1 Cheng's generalization

By this point, most of Cheng's generalization has already been established. Section 5 identified exactly those SVs whose transitive uses form telic VPs with a quantized object, as diagnosed by compatibility with frame *in*-adverbials: TS-oriented activity and change-of-state SVs, but not pure activity SVs.¹⁸ Sections 3.2 and 3.4 showed that the two classes forming telic transitive VPs admit intransitive alternants. To complete Cheng's generalization, it remains only to show that pure activity SVs, such as *mō* 'pet' or *chuí* 'hammer', fail to alternate. The contrast between (86) and (87) confirms that this is indeed the case, providing further support for one of the paper's central claims: not all activity SVs are alike in Mandarin.

- (86) a. #Nèi-zhī māo mō le/ zhèngzài mō. (pure activity SV)
that-CL cat caress PERF PROG caress
Intended: 'The cat has petted/is petting.'

¹⁸Setting aside two classes of verbs not discussed here but that also satisfy Cheng's generalization, namely transitive achievement verbs and verbs of creation and consumption (see Footnote 20).

- b. #*Nèi-ge páizi chuí le/ zhèngzài chuí.* (pure activity SV)
 that-CL plate hammer PERF PROG hammer
 Intended: ‘The plate has hammered/is hammering.’
- c. #*Nèi-liàng chē tuī le.* (pure activity SV)
 that-CL car push PERF
 Intended: ‘The car has pushed.’ (Cheng 1989)
- (87) a. *Nèi-piàn shùlín shāo le/ zhèngzài shāo.* (gradable change-of-state SV)
 that-CL wood burn PFV/PERF PROG burn
 ‘The wood (has) burned/is burning.’
- b. *Nèi-jàn shàngyī xǐ le.* (TS-oriented activity SV)
 that-CL coat wash PERF
 ‘The coat has washed.’

The obvious question is why Cheng’s correlation holds. On the face of it, the aspectual structure of a transitive VP and the availability of an intransitive alternant are logically independent: the former concerns the aspectual structure of the VP, the latter its argument structure. A first clue comes from the interpretation of the intransitive itself. A Mandarin intransitive with an internal argument in subject position invariably predicates a state of the theme. With change-of-state SVs, this is the lexically encoded result state (in (87a), that the wood is burnt).

With TS-oriented activity SVs, too, the intransitive construal depends on a currently relevant state of the theme, made available by sentence-final *le*, as we argue below. The next subsection develops this insight into an explanation of Cheng’s generalization.

6.2 Explaining Cheng’s generalization and the role of sentence-final *le*

We begin by reformulating the core of Cheng’s generalization as the following Mandarin lexical restriction, in the spirit of the restriction proposed by [Kratzer \(2000\)](#), [Rapp \(2015\)](#) and [Gehrke \(2015\)](#) for the German adjectival passive:

- (88) LEXICAL RESTRICTION ON THE MANDARIN TRANSITIVE-INTRANSITIVE ALTERNATION. Only transitive verbs that can form telic VPs with a quantized object can be used intransitively in a default context (i.e., without resorting to some pragmatic licensing).

The question is what underlies this restriction. Building on the insight developed in the previous subsection, we argue that it follows from a more general state-property requirement:

- (90) STATE PROPERTY REQUIREMENT ON MANDARIN INTRANSITIVE CONSTRUALS. An intransitive construal with an internal argument in subject position is felicitous only if the event denoted by the predicate provides evidence for a state property of the internal argument.

The State Property Requirement makes an evidential relation between the event and a state of the internal argument the licensing condition on the intransitive construal. Cheng’s generalization follows if the state property for which the event provides evidence in the intransitive construal also underlies the telicity of the corresponding transitive VP. We argue that this is indeed the case. The source of the relevant state property differs across the three classes—lexical, conventional, or contextual—but whenever such a property licenses the intransitive construal, it also supplies or

helps identify the telos of the corresponding transitive VP.

Change-of-state SVs represent the canonical case covered by the State Property Requirement in (90). They lexically encode a result state, so the event denoted by the predicate directly provides evidence for the corresponding state of the theme, thereby satisfying the State Property Requirement licensing the intransitive construal. The same result state determines the telos of the corresponding transitive VP. Consequently, change-of-state SVs form telic transitive predicates and freely admit intransitive alternants (with or without sentence-final *le*).

The second class comprises verbs that do not lexically encode a result state but are conventionally associated with one, including TS-oriented activity verbs such as *xǐ* ‘wash’ and creation and consumption verbs such as *xiě* ‘write’, *chī* ‘eat’. Unlike change-of-state SVs, these are non-scalar (the scale needed for telicity being provided by the incremental theme rather than by a lexical scale of change; see, e.g., Rappaport Hovav 2014: section 12.6), and non-result verbs. They are, however, associated with a state property. TS-oriented activity verbs such as *xǐ* ‘wash’ are conventionally associated with the theme’s being cleaner, while creation and consumption verbs are associated with the theme’s coming into existence or being consumed. This association helps identify a natural endpoint for the event, namely the point at which the relevant activity has been carried out in full, which, for TS-oriented activity SVs, normally coincides with the conventionally associated state becoming true. This accounts for their telicity.

We argue, on the basis of the distribution of perfect (sentence-final) *le*, that a conventionally associated state alone is not sufficient for the condition in (90) to be met by activity SVs. The associated state must also become the currently relevant state property predicated of the theme. We propose that this is precisely the role of sentence-final *le*.

Following Li and Thompson (1981), we analyze sentence-final *le* as a stativizing perfect signaling a currently relevant state.¹⁹ We formalize this proposal by adopting Piñón (2014) analysis of the perfect, which builds on Nishiyama and Koenig’s (2010) stative analysis by introducing an evidential relation between an event and a set $\mathcal{S}$ of contextually recoverable properties of states (Q-properties):

(91) Representation of the perfect (Piñón 2014):

$$\begin{aligned} &\lambda\mathcal{S}.\exists e(P(e) \wedge \tau(e) \prec t_t) \wedge \\ &\quad \exists e(P(e) \wedge \tau(e) \prec t_t \rightarrow \mathcal{S} \wedge \\ &\quad \mathcal{S} = \lambda Q.\exists s(Q(s) \wedge \tau(s) \circ t_t)) \end{aligned}$$

‘a predicate of sets $\mathcal{S}$ of properties of states such that there is an eventuality e of type P (the event predicate) before the topic time t_t and the fact that there is an eventuality of this type before t_t is evidence for $\mathcal{S}$, where $\mathcal{S}$ is the set of properties Q of states such that there is a state s of type Q and s overlaps temporally with t_t .’

Informally, Piñón’s analysis says that there is an event e before the topic time (t_t), and the fact that there is such an event constitutes evidence for a set $\mathcal{S}$ of state properties holding at the topic time. More precisely, $\mathcal{S}$ is the set of Q-properties true of states holding at the topic time. For present purposes, the crucial feature of this analysis is precisely this evidential relation: the perfect requires a state property for which the event provides evidence.

Our proposal is that, for TS-oriented activity verbs, the conventionally associated state sup-

¹⁹Not all analysis of the perfect takes it to yield a state. On Extended-Now (XN) approaches (McCoard 1978, a.o.), the perfect locates the underlying eventuality within an interval abutting the reference time rather than introducing a state.

plies exactly the Q-property required by the perfect's evidential relation. Possible Q-properties corresponding to the conventionally associated states of TS-oriented activity verbs such as *xǐ* 'wash' or *zhǔ* 'cook' are given in (92).

- (92) a. λs .the-garment-be-clean(er)(s), ...
 b. λs .the-food-be-ready-to-eat(s), ...

Under Piñón's analysis, *le* establishes an evidential relation between, for example, a washing event and the corresponding Q-property in (92a). We propose that, by doing so, it makes this Q-property the currently relevant state property predicated of the theme. As shown in (91), antiagentive construals are acceptable, in a default context, only in its presence. By contrast, the progressive, which does not establish such an evidential relation, is strongly degraded with antiagentives.

- (93) a. *Nèi-ge píngguǒ chī le/ #Nèi-ge píngguǒ zài chī.*
 that-CL apple eat PERF that-CL apple PROG eat.
 'The apple has eaten/ is eating.'
 b. *Yīfu xǐ le/ #Yīfu zài xǐ.*
 coat wash PERF coat PROG wash
 'The coat has washed/ is washing.'

Change-of-state SVs behave differently. Because their state property is lexically encoded, it is already available for predication of the theme. The event denoted by the predicate therefore directly provides evidence for the corresponding state property, so the condition in (90) is met independently of sentence-final *le*. Consequently, change-of-state SVs form intransitive alternants with or without *le* and freely combine with the progressive (cf. 87a)).

The role of conventionally associated states can thus be stated more precisely. First, they help identify a natural endpoint for the transitive activity description, thereby accounting for the telicity of the corresponding transitive VP. Second, under Piñón's analysis, they supply the relevant Q-property for which the event provides evidence, thereby meeting the State Property Requirement on the intransitive construal. This explains why these verbs form telic transitive predicates and admit intransitive alternants, while, unlike change-of-state SVs, requiring sentence-final *le* in the intransitive use.²⁰

The third class in our three-way typology comprises pure activity verbs such *mō* 'pet' or *chàng* 'sing'. Unlike the preceding two classes, they neither lexically encode nor are conventionally associated with a state of the theme. Consequently, in a default context, an event of petting or singing does not provide evidence for any particular currently relevant state of the theme. The State Property Requirement is therefore not met, and these verbs resist intransitive construals. Correspondingly, their transitive uses lack a natural telos and are atelic.

The present proposal does not predict that such verbs can never receive telic or intransitive construals. Rather, it predicts that both should become available under the same conditions,

²⁰Creation and consumption verbs satisfy the State Property Requirement in the same way as TS-oriented activity verbs, in that sentence-final *le* is required to make their associated state property—the theme's having come into existence or having been consumed—the currently relevant property predicated of the theme. They therefore also fall under Cheng's generalization. They nevertheless differ from TS-oriented activity verbs in important respects. Aspectually, the coming into existence of the effected object and the progressive disappearance of the consumed object unfold in strict tandem with the verbal event, yielding an incremental-theme relation (Krifka 1989). Consequently, unlike TS-oriented activity verbs, they do not license zero-change readings (Tatevosov 2008; Martin and Demirdache 2020). Moreover, the relevant state is not a target state in Parsons's (1990) sense, since it is neither reversible, nor transitory.

namely when the context is sufficiently enriched to supply the missing state property. This prediction is borne out.

Although atelic VPs such as *mō māo* ‘pet the cat’ are infelicitous in intransitive frames out of the blue, they become acceptable in the ‘job-is-done’ context of the type identified by Kratzer (2000), for instance where it is the speaker’s job to pet the cat. The same context also licenses the corresponding telic transitive construal (as in German; see Rapp 2015: 510), as evidenced by the acceptability of the frame adverbial in (94).

- (94) *Lùlu liǎng fēnzhōng nèi jiù mō le nèi-zhī māo.* OK in the *job-is-done* context
 Lulu two minute within JIU caress PFV that-CL cat
 ‘Lulu petted the cat in two minutes.’

The antiagentive construal, however, requires an additional ingredient: sentence-final *le* together with a contextually salient state that the theme comes to hold as a result of the event. This state may simply be the resultant state in Parsons’s (1990) sense: the state of having been the theme of the verbal event. Thus, the Mandarin example in (95) is felicitous when used to convey that the cat is already petted. It is equally felicitous to convey that the cat is now calm and settled, its earlier restlessness having been soothed away by the stroking.

- (95) *Māo mō le.*
 cat pet PERF
 ‘The cat has petted.’

Under the present proposal, this enriched context crucially supplies the relevant state property. Possible contextually supplied *Q*-properties for (95) are given in (96).

- (96) a. λs .the-cat-be-petted(*s*)
 b. λs .the-cat-be-calm-and-settled(*s*)
 c. λs .the-cat-be-soothed(*s*)
 d. ...

Unlike TS-oriented activity verbs, whose relevant *Q*-property is conventionally associated with the verb, the relevant *Q*-property here is supplied entirely by contextual enrichment. Sentence-final *le* can therefore establish the required evidential relation only when the enriched context supplies a sufficiently salient *Q*-property for which the event provides evidence.

Cheng’s generalization thus follows from a single underlying principle. Across all three classes of SVs, the same state property for which the event provides evidence in the intransitive construal also supplies or helps identify the telos of the corresponding transitive VP. What distinguishes the three classes is simply how that state property is identified: lexically, conventionally, or contextually. The way in which these three sources of the state property align with telicity and the transitive–intransitive alternation is summarized in Table 2.

7 Conclusion

A long-standing controversy in the literature concerns the lexical semantics of Mandarin simple verbs corresponding to English causative verbs. Because verbs such as *shāo* ‘burn’ permit zero-change construals with an animate agent, they have widely been assumed not to be genuine change-of-state predicates and have instead been assimilated to verbs such as Mandarin *xǐ* and English *wash*—Talmy’s (2000) implied-fulfillment verbs, which describe an activity directed toward

	source of the state property	telic transitive VP	intransitive alternant	final <i>le</i> required
Change-of-state SVs (<i>shāo</i> , <i>shā</i>)	lexical	✓	anticausative	no
TS-oriented activity SVs (<i>xǐ</i>)	conventional	✓	antiagentive	yes
Pure activity SVs (<i>chuí</i> , <i>mō</i>)	contextual	#	#	yes

Table 2: The source of the theme’s state property predicts both the telicity of the transitive VP and the availability and type of the corresponding intransitive alternant (✓ = available; # = unavailable out of the blue).

bringing about a target state without entailing that it is attained. True change-of-state predicates are consequently assumed to be expressed only by resultative compounds.

We argue against this view by showing that the distribution of zero-change construals reveals that these verbs do not form a lexically homogeneous class. Although both *shāo* ‘burn’ and TS-oriented activity verbs such as *xǐ* ‘wash’ permit zero-change construals in transitive clauses with an agent subject, their behavior across syntactic frames and aspectual environments reveals a fundamental lexical distinction: verbs such as *shāo* ‘burn’ lexically encode a result state. Unlike TS-oriented activity verbs, they reject zero-change construals in intransitive frames and with causer subjects, and their transitive uses lose the zero-change interpretation under completive marking. Mandarin therefore possesses underived simple verbs with genuine change-of-state semantics alongside TS-oriented activity verbs.

The apparent activity-like behavior of gradable change-of-state SVs receives a different explanation. We argued that Mandarin *shāo* ‘burn’ and English *burn* encode different kinds of degree: the former a degree of *event realization* d_e (Piñón 2008), the latter a degree of *change* d_c (Kennedy and Levin 2008). When d_e receives a non-maximal value under Agent Voice, the event may be realized to a positive degree without the theme undergoing any change. The apparent activity-like behavior of Mandarin *shāo* therefore follows not from the absence of a lexically encoded result state, but from the interaction between degree semantics, argument structure and Voice. More generally, this analysis identifies a previously unrecognized source of crosslinguistic variation among gradable change-of-state predicates.

A second consequence of this lexical distinction is a corresponding distinction between two superficially similar intransitive alternations. Change-of-state verbs undergo an CAUSATIVE / ANTI-CAUSATIVE alternation, whereas TS-oriented activity verbs undergo a distinct AGENTIVE / ANTI-AGENTIVE alternation. Here too, the distribution of zero-change construals provides the crucial diagnostic: whereas the antiagentive alternant tolerates a zero-change construal, the anticausative alternant does not. We provide the first formal semantic analysis of the antiagentive alternation: although antiagentives retain the semantic ACT predicate contributed by the v_{do} head, and hence the conceptual presence of an agent, they lack a syntactic Voice projection, leaving the theme as the sole syntactically projected argument. Their incompatibility with agent-oriented adverbials and instrumental phrases, together with their incompatibility with *zìjǐ* ‘by itself’, distinguishes antiagentives from passives and middles. Although first identified as such here for Mandarin, comparable antiagentive alternations have been reported in Salish languages (Davis and Demirdache 2000), Hindi (Bhatt and Embick 2017), Brazilian Portuguese (Carvalho 2016), and Puyuma

(Chen 2022).

A further contribution is the demonstration that genuine activity verbs do not form a homogeneous class. We distinguished TS-oriented activity verbs, which are conventionally associated with a state property of the theme, from pure activity verbs, which lack such an association. Here again, the distribution of the zero-change construal proves decisive. Whereas TS-oriented activity verbs preserve the zero-change construal in their antiagentive alternants, pure activity verbs do not alternate at all. Accordingly, like change-of-state verbs, only TS-oriented activity verbs—but not pure activity verbs—fall under Cheng’s (1989) generalization that precisely those SVs whose transitive uses form telic predicates with a quantized object admit an intransitive alternant.

Our explanation of Cheng’s (1989) generalization rests on a single principle: the state property for which the event provides evidence in the intransitive construal also underlies the telicity of the corresponding transitive VP. The distinctions established in this paper yield a three-way typology of Mandarin SVs—change-of-state, TS-oriented activity, and pure activity verbs—that exactly matches Cheng’s generalization. The typology reflects three distinct ways in which the relevant state property may be identified: it may be lexically encoded, conventionally associated with the verbal meaning, or, in marked cases, supplied by contextual enrichment.

Finally, we contribute to a more differentiated theory of non-culmination by showing that superficially similar zero-change interpretations in Mandarin arise from two distinct semantic sources. For gradable change-of-state verbs, non-culmination follows from a non-maximal interpretation of the degree of event realization under Agent Voice; for non-gradable change-of-state verbs, it follows from the semantics of Mandarin *le* as a weak partitive perfective operator. The agentivity requirement and the cardinality adverbial effect emerge as diagnostics of these two semantic sources rather than as independent stipulations. More broadly, by explaining how the intricate pattern of distribution of zero-change construals reflects underlying differences in lexical semantics, argument structure, and aspectual composition, we establish the zero-change construal as a novel probe into event structure.

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